FOURTH YEAR PROJECT DOCUMENTATION

MKULIMAHUB: A WEB-BASED SYSTEM FOR SECURE REAL-TIME AGRICULTURAL ADVICE ACCESS IN KENYA

by

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A RESEARCH DOCUMENTATION SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE, SCHOOL OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF AN AWARD OF A DEGREE IN BACHELOR OF SCIENCE COMPUTER SCIENCE OF DEDAN KIMATHI UNIVERSITY

JANUARY, 2026

DECLARATION

I hereby declare that this project is my original work and to the best of my knowledge the project has not formed the basis of any other award.

Admission No: C026-01-0997/2021

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Supervisor

This report has been submitted for examination with my approval as the University Supervisor

Dr. George Musumba

Signature ……………… Date……………………

ABSTRACT

In Kenya, a significant deficit of agricultural extension services exists, with 1 farmer to 3,000 extension officers, compared with the FAO recommended ratio of 1:400. This gap leaves farmers at the mercy of unverified sources of information and leads to post-harvest losses of 20-30 percent per year, and huge financial losses. The project developed Mkulima Hub, a web-based platform that integrates an AI-powered chatbot with the Gemini API, verified expert consultation with M-Pesa payment integration, localized stage-by-stage crop guides, moderated community forum, and real-time weather data with a five-day forecast. The system was developed using MERN stack (MongoDB, Express.js, React, Node.js), Design Science Research methodology and Agile Scrum framework. The problem severity was validated using a market validation survey of 12 respondents across five counties in Kenya. The survey showed that 33.3% of the farmers lost more than KES 20,000 in the past year as a result of wrong agricultural advice. Another 25.0% lost between KES 5,000 and 20,000. The functional platform was successfully deployed at www.mkulimahub.vercel.app and provides three role-based dashboards for farmers, agricultural experts and administrators. Project highlights included: A Gemini powered chatbot that understands Swahili and English farming queries, A functional consultation booking engine that included real time video and chat, an administrative interface that incorporated user management, transaction tracking, system logs and analytics. “On the platform it was shown that an integrated digital ecosystem can fill the gap in the agricultural extension service. Future works recommendation: offline PWA features, USSD integration for feature phone users, expansion to livestock management and agri-finance services.

DEDICATION

I dedicate this project to Almighty God for His grace and guidance throughout my academic journey.

To my parents, for their constant support, sacrifices and encouragement which have been my foundation and strength. You have never wavered in your belief in me.

This platform is for all smallholder farmers in Kenya who work tirelessly to feed our nation. We hope this works for you and improves your livelihoods.

ACKNOWLEDGMENT

I thank Almighty God for good health and strength throughout this project.

I am sincerely grateful to my supervisor Dr. Musumba for his invaluable guidance, patience and mentorship from proposal to completion.

I appreciate my lecturers in the Department of Computer Science, Dedan Kimathi University of Technology, for equipping me with skills that made this work possible.

I am grateful to the farmers, the agricultural scientists and the administrators who participated in the survey and pilot testing. Your time and sincere feedback gave practical significance to this research.

I am grateful to my family for their constant financial and emotional support.

Finally, I would like to thank my friends and classmates for their support and teamwork.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DFD	Data Flow Diagram
DSR	Design Science Research
FAO	Food and Agriculture Organization
GDP	Gross Domestic Product
GPS	Global Positioning System
IoT	Internet of Things
JWT	JSON Web Token
KALRO	Kenya Agricultural and Livestock Research Organization
LLM	Large Language Model
MERN	MongoDB, Express.js, React, Node.js
MVP	Minimum Viable Product
NGO	Non-Governmental Organization
NLP	Natural Language Processing
PWA	Progressive Web Application
REST	Representational State Transfer
SMS	Short Message Service
UAT	User Acceptance Testing
UML	Unified Modeling Language
UI/UX	User Interface / User Experience

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Agriculture remains the backbone of Kenya's economy, contributing approximately 33 percent of the Gross Domestic Product (GDP) and employing over 40 percent of the population, with smallholder farmers accounting for nearly 75 percent of total agricultural output (Ministry of Agriculture, 2022). Despite this critical role, smallholder farmers in Kenya have long been underserved by the traditional agricultural extension system, which was designed for a different era with a smaller farming population. The Food and Agriculture Organization (FAO) recommends a farmer-to-extension-officer ratio of 1:400 for effective service delivery. In stark contrast, Kenya currently operates at a ratio of approximately 1:3,000, creating a profound service vacuum that leaves millions of farmers without access to timely, localized, and expert advisory services (FAO, 2021).

This systemic gap has forced farmers to seek agricultural information from alternative sources, including fellow farmers, social media groups, agrovet salespeople, and radio programs. While some of these sources provide useful guidance, much of the information is unverified, contextually inappropriate, or outright incorrect. The consequences of this information deficit are severe and quantifiable. Post-harvest losses in Kenya are estimated at 20 to 30 percent annually for staple crops such as maize, beans, and vegetables, directly attributed to poor harvesting techniques, inadequate pest and disease management, and suboptimal storage practices (KALRO, 2021).

The global digital transformation of agriculture has demonstrated significant potential to address such challenges. In developed economies, precision agriculture tools, data-driven advisory platforms, and AI-powered decision support systems have improved yields, reduced input costs, and enhanced sustainability. In developing contexts, mobile-based agricultural services such as WeFarm (peer-to-peer knowledge sharing), iCow (livestock and crop management), and Digital Green (video-based extension) have shown promise in reaching previously underserved farming communities. However, these solutions remain fragmented, each addressing only one aspect of a complex problem. No single platform currently exists that integrates AI-powered preliminary guidance, verified expert consultation, localized crop management information, moderated community support, and real-time weather intelligence into a unified digital ecosystem tailored for Kenyan smallholder farmers.

1.2 Statement of the Problem

Smallholder farmers in Kenya faced a profound challenge in accessing reliable, timely, and localized agricultural advisory services, leading to suboptimal farming practices, high post-harvest losses, and diminished incomes. The core of this problem was the severe shortage of government extension officers, creating a service vacuum where the farmer-to-extension-agent ratio stood at approximately 1:3,000, far below the FAO-recommended ratio of 1:400 (Ministry of Agriculture, 2022). This deficit forced farmers to seek agricultural information from alternative and often unreliable sources.

Consequently, inadequate pest and disease management resulting from limited expert guidance contributed to post-harvest losses estimated at 20 to 30 percent annually, significantly affecting household food security and national economic output (Food and Agriculture Organization [FAO], 2021). Furthermore, the widespread circulation of unverified information through social media platforms and informal peer networks exacerbated the problem, frequently leading to financial losses due to the adoption of ineffective or harmful farm inputs (FAO, 2021).

While several digital agricultural solutions such as WeFarm, which focused on peer-to-peer knowledge sharing, and Twiga Foods, which primarily addressed market linkages, had emerged, the digital agriculture landscape remained fragmented (Omulo & Kumeh, 2020; GAFSP, n.d.). There was currently no unified platform that integrated verified expert consultation, AI-powered preliminary guidance, localized crop management information, and community-based farmer support into a single, accessible ecosystem tailored for Kenyan smallholder farmers. This fragmentation represented a significant barrier to agricultural modernization, productivity enhancement, and long-term sector resilience.

To address this gap, this project developed Mkulima Hub, a web-based platform that consolidated these essential services into one integrated ecosystem, thereby providing smallholder farmers with reliable, on-demand access to verified agricultural knowledge and expert support.

1.3 Objectives

1.3.1 General Objective

To design, develop, and deploy an integrated digital platform (Mkulima Hub) that connects smallholder farmers in Kenya with verified agricultural experts and localized knowledge to improve farming productivity and decision-making.

1.3.2 Specific Objectives

a) To develop an AI-Powered ChatBot system that understands local farming terminology and instantly connects farmers to solutions.
b) To create localized, stage-by-stage crop guides tailored to specific agro-ecological zones.
c) To implement a secure expert consultation platform with integrated M-Pesa payment processing and real-time video/text communication.
d) To develop a moderated community forum where experts ensure the quality and accuracy of peer-to-peer information.
e) To integrate real-time weather data and 5-day forecasting to support climate-smart agricultural decisions.
f) To evaluate the platform's usability, relevance, and potential impact through a pilot deployment and collection of user feedback.

1.4 Research Questions

The following research questions guided this study:

i. What were the primary sources of agricultural information for smallholder farmers, and what were their major unmet needs regarding expert advice and crop management?

ii. How could a digital platform be architecturally designed to effectively facilitate seamless interaction between farmers, verified experts, and administrators?

iii. What technical integrations were required to create a cohesive user experience encompassing AI assistance, crop guides, expert consultation, and community features?

iv. What was the perceived usability, usefulness, and willingness-to-pay among target users during the pilot phase of the Mkulima Hub platform?

1.5 Justification of the Study

This research was justified by its potential to generate significant socio-economic impact. The Mkulima Hub platform directly addressed a critical market and service failure in Kenya's agricultural sector. The project benefited multiple stakeholders: smallholder farmers gained affordable, on-demand access to credible expertise, potentially increasing their yields and incomes; agricultural experts obtained a new channel to monetize their knowledge and extend their reach; and policymakers and agricultural organizations received valuable, data-driven insights into farmer needs and challenges. Furthermore, by promoting better farming practices, the project contributed to national goals of food security, climate resilience, and rural development. For the researcher, this project provided a practical application of software engineering, user-centered design, and system integration skills to solve a real-world problem.

1.6 Scope of the Study

The scope of this research is limited to the idea generation, design, and preliminary testing of the Mkulima Hub platform. The major theme of the investigation was offering advisory services concerning the cultivation of crops, focusing on major crops important for farmers in Kenya.

The target population comprised small-scale farmers, certified agricultural specialists (such as agronomists and extension workers), and platform administrators. This investigation covers the software development life cycle starting with requirements elicitation and ending up at MVP implementation and testing. It does not include the business model evaluation and scaling process as well as hardware acquisition for farmers.

1.7 Limitation of the Study

The implementation of this study faced certain limitations. Firstly, the reliance on self-reported data from the farmer survey (Appendix A) may have introduced biases. Secondly, the digital literacy levels of the target farmer population may have affected the rate of adoption and the quality of feedback during testing. Thirdly, the success of the expert consultation feature was contingent on the project's ability to onboard and retain a critical mass of qualified experts, which was challenging initially. Finally, the performance and relevance of the AI chatbot were dependent on the quality and specificity of the training data collected, which remained an ongoing process. These limitations were acknowledged, and their mitigation strategies were discussed in the final project report.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter presented a critical analysis of existing literature, case studies, and digital solutions relevant to the problem of agricultural knowledge dissemination and the use of Information and Communication Technology (ICT) in farming. The purpose was to establish a theoretical foundation for the Mkulima Hub project by examining what had been done, identifying effective methodologies, highlighting gaps, and positioning the proposed solution within the current landscape. The review was structured around key themes: the global and local state of digital agriculture, analysis of specific platforms, and the technological frameworks used in similar interventions.

2.2 Case Studies of Existing Digital Agricultural Platforms

2.2.1 Case Study 1: WeFarm – A Peer-to-Peer Knowledge Sharing Network

WeFarm was a global peer-to-peer digital platform that enabled small-scale farmers to ask agricultural questions and receive responses from fellow farmers via SMS or web-based interfaces without requiring internet connectivity. The platform employed a crowdsourcing methodology in which farmer-generated queries were distributed across the network and answered by peers.

Research by Wyche and Steinfield (2016) highlighted WeFarm's effectiveness in reaching offline and resource-constrained populations while fostering community-based problem-solving and experiential knowledge exchange. However, a critical limitation identified in the literature was the absence of expert verification mechanisms. While responses were timely and contextually relevant, they were not validated by agricultural professionals, creating a risk of misinformation dissemination. Mkulima Hub addressed this limitation by incorporating a moderated expert verification layer that ensured credibility while preserving community interaction.

2.2.2 Case Study 2: iCow – A Mobile-Based Livestock and Crop Management Tool

iCow was a Kenyan mobile agricultural platform that delivered voice- and SMS-based advisory services on livestock management, crop production, and market information. The system operated using a structured content delivery model where predefined agricultural guidance was periodically pushed to subscribed farmers.

Krell et al. (2021) commended iCow for its localized content, accessibility on basic mobile phones, and effectiveness in improving farm management practices. Despite these strengths, the platform's primary limitation lay in its static, one-directional communication model. Farmers were largely passive recipients of information with limited ability to seek personalized, real-time responses. In contrast, Mkulima Hub provided a two-way, interactive advisory system that enabled farmers to ask context-specific questions and receive tailored guidance.

2.2.3 Case Study 3: Digital Green – A Video-Based Agricultural Extension Approach

Digital Green employed a participatory, community-centered approach to agricultural extension by producing and disseminating locally relevant instructional videos through facilitated farmer groups. The platform integrated digital media with human intermediaries, emphasizing social learning and peer influence.

Empirical studies by Gandhi et al. (2019) demonstrated that Digital Green significantly improved adoption rates of recommended farming practices due to its localized and socially embedded dissemination model. However, a notable limitation was its asynchronous and non-interactive nature. While effective for demonstration and awareness creation, it did not support real-time question-and-answer interactions or address individual farmer-specific challenges. Mkulima Hub built upon the strength of localized content while extending functionality through synchronous expert consultations and interactive support.

2.2.4 Case Study 4: AgriChatbox (South Africa) – An AI-Powered Conversational Agent

AgriChatbox was an AI-driven agricultural chatbot deployed in South Africa, designed to respond to farmer inquiries through natural language interfaces on popular messaging platforms. The system utilized natural language processing and a knowledge base curated by agricultural experts.

Van der Spuy and van Biljon (2022) reported that AgriChatbox successfully provided instant, scalable first-line agricultural support, particularly for frequently asked questions. However, challenges identified included the "black box" nature of AI decision-making, limitations in handling complex or highly localized scenarios, and insufficient escalation pathways to human experts. Mkulima Hub adopted a hybrid advisory model in which artificial intelligence served as the initial interaction layer, with deliberate and seamless escalation to verified human experts for complex, high-risk, or ambiguous cases.

2.3 Summary

The reviewed literature and case studies revealed a consistent evolution from basic information dissemination towards more interactive, intelligent, and user-centric platforms. Key strengths in existing solutions included the use of low-tech channels for reach (SMS), community-driven content, and the emerging use of artificial intelligence for scalability. However, critical weaknesses were identified:

a. Credibility versus Crowdsourcing Trade-off: Platforms like WeFarm offered scale through community but lacked expert validation.
b. Interactivity Limitation: Tools like iCow and Digital Green were effective for broadcasting information but were not designed for personalized, interactive problem-solving.
c. Fragmentation of Services: Most platforms specialized in one aspect (e.g., peer advice, market prices, static guides) rather than offering a unified ecosystem.
d. AI Reliability and Escalation: AI-only solutions could fail on nuanced issues and often lacked a clear pathway to human expert intervention.

These weaknesses created a clear opportunity for an integrated solution that combined verified expertise, intelligent automation, interactive communication, and community features in a single platform.

2.4 Research Gap

Despite the proliferation of digital agriculture tools, there was a discernible gap in the Kenyan market for a single, integrated platform that seamlessly combined:

- An intelligent, context-aware artificial intelligence assistant.
- Guaranteed escalation to verified, locatable agricultural experts.
- A repository of localized, stage-by-stage crop guides.
- A quality-moderated peer community.
- Real-time weather data and five-day predictions.
- Integrated mobile money payments (M-Pesa).

No existing solution examined provided this holistic, multi-layered support system.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter presented the process that was followed while coming up with the Mkulima Hub software application. The research approach, methods used in collecting data, software development technique used, system architecture, and the method applied for testing the system are all covered in this chapter. All this was done with the aim of creating an action plan on how the research objectives would be achieved.

3.2 Research Design

The current study adopted the Design Science Research (DSR) approach. DSR was guided by the development and evaluation of artifacts aimed at addressing the real-life challenges encountered. These artifacts were created through a series of activities which included identifying problems, defining objectives, designing and developing artifacts, demonstrating, and evaluating them. The DSR model worked perfectly well alongside the Agile development methodology since each iteration of the platform had to be technically and functionally robust.

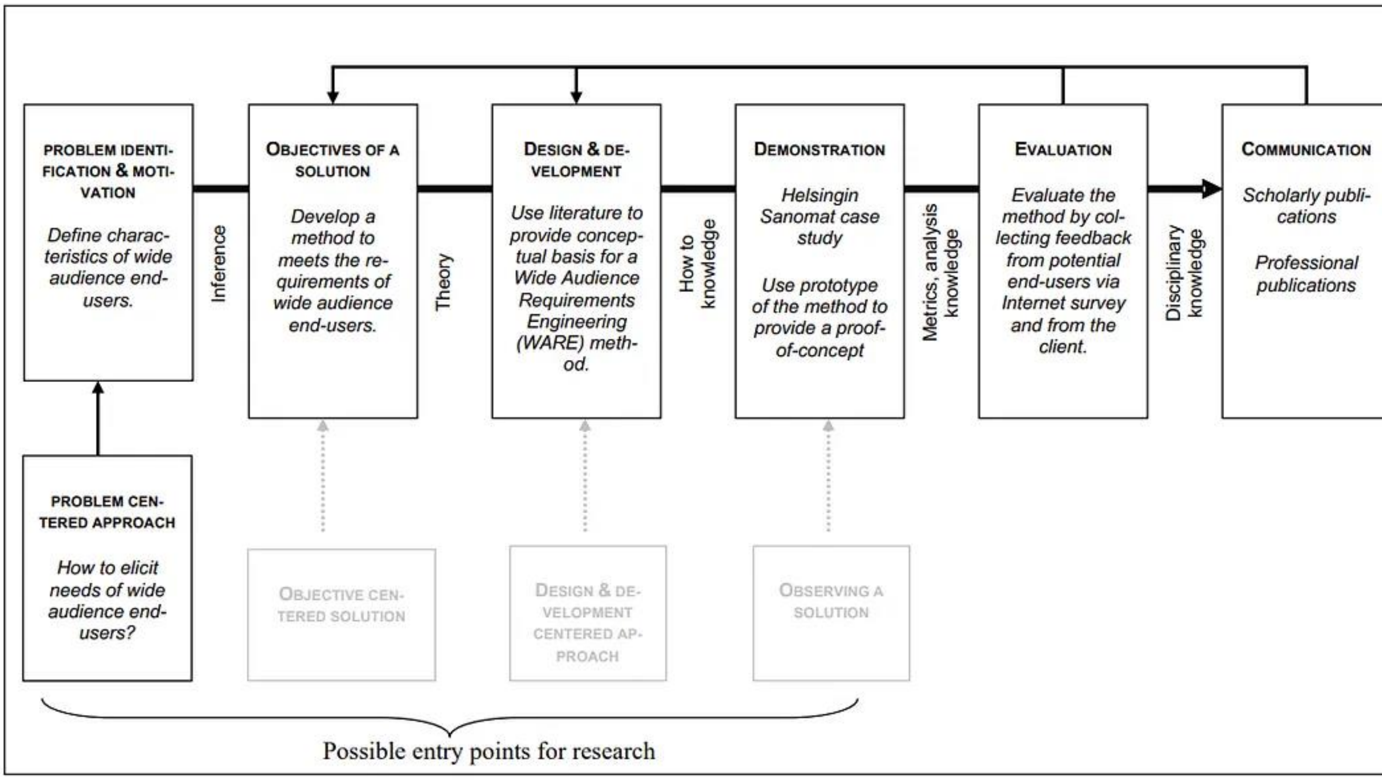

Figure 1 DSR process case study | Reference: Peffers et al (2007)

3.3 Data Collection and Fact-Finding Techniques

To ensure that the solution was based on user requirements and realities, the following methods were employed:

Technique	Description
Quantitative Survey	This involved an online survey questionnaire, which is shown in **Appendix A**, designed to validate the market and assess user needs. The questionnaire, administered through the website portal, gathered information regarding user demographics, problem severity, current information sources, and feature demands. www.mhtool.vercel.app
Analysis of Secondary Data	Existing agricultural databases, KALRO publications, and FAO publications were analyzed to provide scientifically accurate and location-specific content for the crop guides and artificial intelligence knowledge base.
Technical Prototyping	Creating a working prototype (www.mkulimahub.vercel.app) provided a form of active research to validate technical feasibility and functionality of the product.

Table 1 Data Collection Techniques

3.4 Software Development Methodology

The Agile Scrum methodology was used for the completion of this project. Agile Scrum is an iterative and incremental approach which provides flexibility, user-centricity, and the ability to deal with uncertainties during software development.

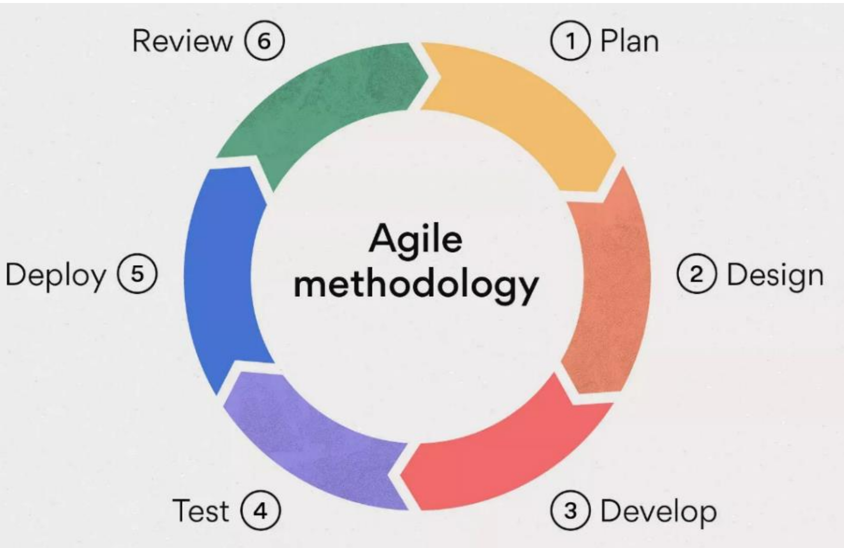

Figure 2 Agile Methodology | Reference: www.asana.com/uses/agile-management

Process: The work was broken up into two-to-three-week sprints using time boxing. At the start of every sprint, there would be a sprint planning session where certain features were chosen from the Product Backlog, which consisted of the prioritized requirements list.

Ceremonies:

Ceremony	Purpose
Daily Stand-ups	Synchronization of activities and blockers (15 minutes every day)
Sprint Review	Successful completion of tasks at the end of each sprint
Sprint Retrospective	Process reflection and identification of areas for improvement for the next sprint

Table 2 Scrum Ceremonies

Conclusion: Every sprint ended with the creation of an increment of a shippable product that could be integrated and tested continuously.

In order to overcome the research gap that was identified, the methodology used for developing Mkulima Hub has resulted in MVP with core features (expert matching, crop guidance, appointment booking) to be developed first, and then further sprints with advanced features (AI chatbot, forum, offline capabilities for the PWA).

3.5 System Design and Architecture

3.5.1 Architectural Style

The application used a client-server architecture with a microservices-based backend for better scalability. It was designed as a Progressive Web App (PWA) to allow usage across various devices The application used a client-server architecture with a microservices-based backend for better scalability. It was designed as a Progressive Web App (PWA) to allow usage across various devices

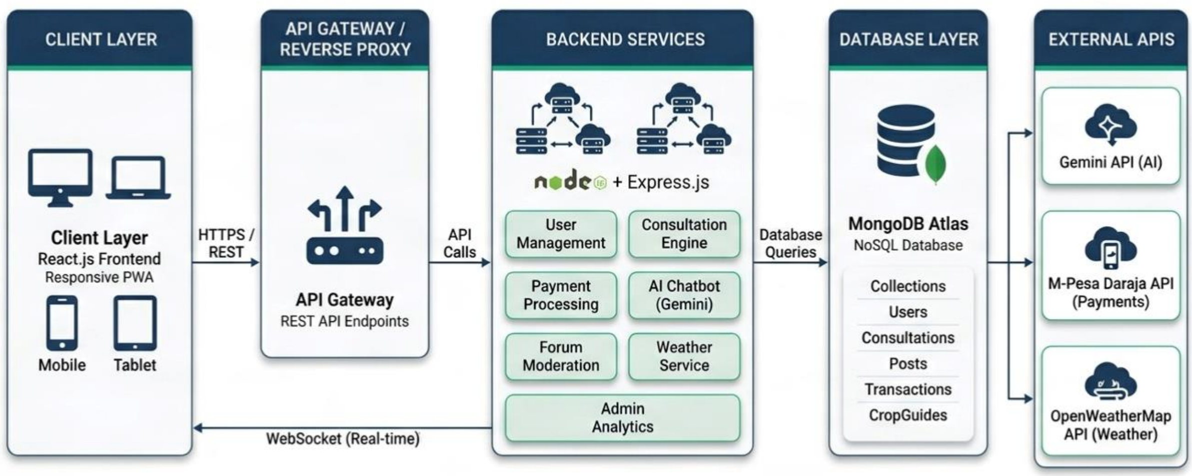

Figure 3 client-server architecture

3.5.2 Technology Stack (MERN):

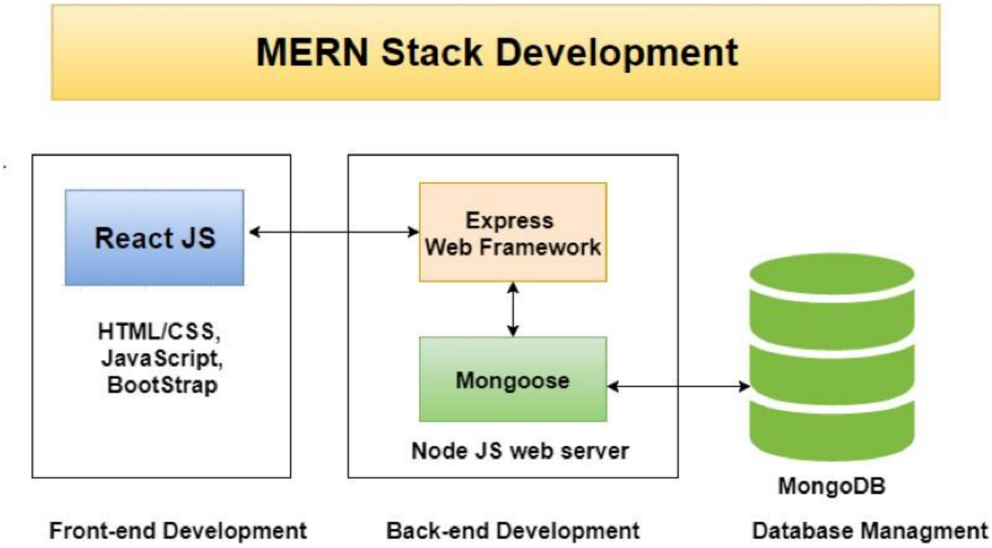

Figure 4 MERN Stack

Component	Technology	Purpose
MongoDB	NoSQL Database	Flexible, JSON-like data storage
Express.js	Backend Framework	RESTful API development for handling business logic and data operations
React.js	Frontend Library	Dynamic, responsive user interface with component-based architecture
Node.js	Runtime Environment	Server-side JavaScript execution for scalable backend services

Table 3 MERN Stack

3.5.3 External API Integrations

API	Purpose
Gemini API	Powered the AI chatbot for natural language understanding in English and Swahili
M-Pesa Daraja API	Facilitated secure mobile money payments for expert consultations
OpenWeather API	Provided real-time weather data and five-day forecasts
Google OAuth API	Enabled social login functionality for users

Table 4 External API Integrations

3.5.4 Key Modules & Components:

Module	Description
User Management	Handled authentication and authorization (role-based)
Knowledge Base Module	Managed the repository of localized, stage-by-stage crop guides and frequently asked questions
Consultation Engine	Managed booking, scheduling, video/chat integration (WebRTC and Socket.io), and M-Pesa payment processing
AI Agent Module	Integrated the Gemini large language model API to answer farmer queries in English and Swahili with expert escalation
Weather Module	Integrated OpenWeather API for current conditions and five-day hyperlocal forecasts
Community Forum Module	Provided a real-time discussion platform with expert moderation, post approval, and upvoting
Admin Dashboard Module	Offered comprehensive interface for user verification, transaction tracking, system logs, and analytics

Table 5 Key Modules & Components

3.6 Testing Strategy

Various levels of tests have been performed for software testing purposes:

i. **Unit Tests (Jest)**

Modules such as calculations and user authorization were validated separately to guarantee their correct work at a micro level within the application.

ii. **Integration Tests (Postman, Supertest)**

These types of tests covered the interaction between modules with API endpoints and third-party services integration such as M-Pesa and Gemini API.

iii. **User Acceptance Test (UAT)**

User acceptance testing was performed in the form of task-oriented exercises and feedback forms by target users in a pilot environment.

iv. **Performance Test (Lighthouse, GTmetrix)**

The performance tests consisted of measuring page load times and responsiveness to optimize the performance of the application.

v. **Security Tests (Manual Review, JWT Validation)**

Input validations and security measures such as JWT validation, hashing passwords using bcrypt have been performed.

3.7 Ethical Considerations

Consideration	Implementation
Informed Consent	All survey participants and pilot testers were informed of the study's purpose and how their data would be used anonymously. Consent was obtained prior to data collection to ensure voluntary participation.
Data Privacy & Security	User data was handled following best practices. Personal information (names, contacts, locations) was not shared without explicit consent. Passwords were securely hashed using bcrypt, and sensitive data was protected from unauthorized access.
Data Anonymization	Survey responses were anonymized before analysis to protect respondent identities and ensure confidentiality.
Expert Compensation Transparency	The payment model for experts was clearly defined, ensuring fair compensation of KES 200–300 per 30-minute consultation based on time and expertise.
Accessibility & Inclusive Design	The platform supported accessibility features such as adjustable font sizes, dark/light theme toggle, and screen reader compatibility to enhance usability for users with different needs.
User Account Control & Autonomy	Users had full control over their accounts, including updating personal information, deactivating accounts, and permanently deleting their data upon request.
Content Moderation & Community Safety	A moderation system was implemented for user-generated content and forum interactions to ensure respectful communication, prevent abuse, and maintain a safe environment.

Table 6 Ethical Considerations

CHAPTER 4: ANALYSIS AND DESIGN

4.1 Introduction

This chapter presented the analysis and design phase of the Mkulima Hub platform. It documented the functional and non-functional requirements gathered from stakeholders, the system modeling using Unified Modeling Language (UML) diagrams, the database design, the user interface design principles, and the system architecture design. The goal was to translate the requirements identified in previous chapters into a concrete technical blueprint that guided the implementation phase.

4.2 Requirements Analysis

4.2.1 Functional Requirements

Functional requirements specified the behaviors and functions the system must perform. These were derived from the survey data, literature review, and stakeholder consultations.

Farmer Requirements:

ID	Requirement Description
FR-F01	The system shall allow farmers to register and create an account
FR-F02	The system shall provide an AI-powered chatbot that is multi-lingual
FR-F03	The system shall display localized, stage-by-stage crop guides
FR-F04	The system shall allow farmers to search, view, and book agricultural experts
FR-F05	The system shall integrate M-Pesa for secure payment of expert consultations
FR-F06	The system shall provide real-time weather data and forecasts
FR-F07	The system shall allow farmers to create, edit, and delete forum posts
FR-F08	The system shall allow farmers to rate and review experts after consultations
FR-F09	The system shall provide a dashboard showing crop progress, consultation history, and payment records
FR-F10	The system shall allow farmers to update their profile and preferences

Expert Requirements:

ID	Requirement Description
FR-E01	The system shall allow agricultural experts to register and login to a dashboard showing consultation requests, earnings, and schedule
FR-E02	The system shall allow experts to accept or decline consultation requests
FR-E03	The system shall provide real-time video and text chat functionality for consultations
FR-E04	The system shall allow experts to moderate forum posts
FR-E05	The system shall allow experts to update their profile, availability, and consultation rates
FR-E06	The system shall track expert earnings and provide a withdrawal mechanism

Admin Requirements:

ID	Requirement Description
FR-A01	The system shall provide an admin dashboard with overview of platform activity
FR-A02	The system shall allow admins to view, verify, suspend, or delete all users
FR-A03	The system shall allow admins to view and monitor all transactions
FR-A04	The system shall provide system logs for debugging and audit purposes
FR-A05	The system shall allow admins to view analytics
FR-A06	The system shall allow admins to manage forum content and resolve disputes

4.2.2 Non-Functional Requirements

ID	Category	Requirement Description
NFR-01	Performance	Page load time shall be under 3 seconds on a 3G connection
NFR-02	Security	All passwords shall be hashed using bcrypt; all API calls shall use JWT authentication
NFR-03	Usability	The system Usability Scale (SUS) score shall be above 70
NFR-04	Availability	The system shall have 99.5% uptime during pilot testing
NFR-05	Scalability	The system architecture shall support up to 10,000 concurrent users
NFR-06	Compatibility	The platform shall be responsive
NFR-07	Maintainability	Code shall be modular, documented

4.3 System Modeling (UML Diagrams)

Unified Modeling Language (UML) diagrams were created to visually represent the system's structure and behavior.

4.3.1 Use Case Diagram

The use case diagram illustrated the interactions between actors and system functionalities.

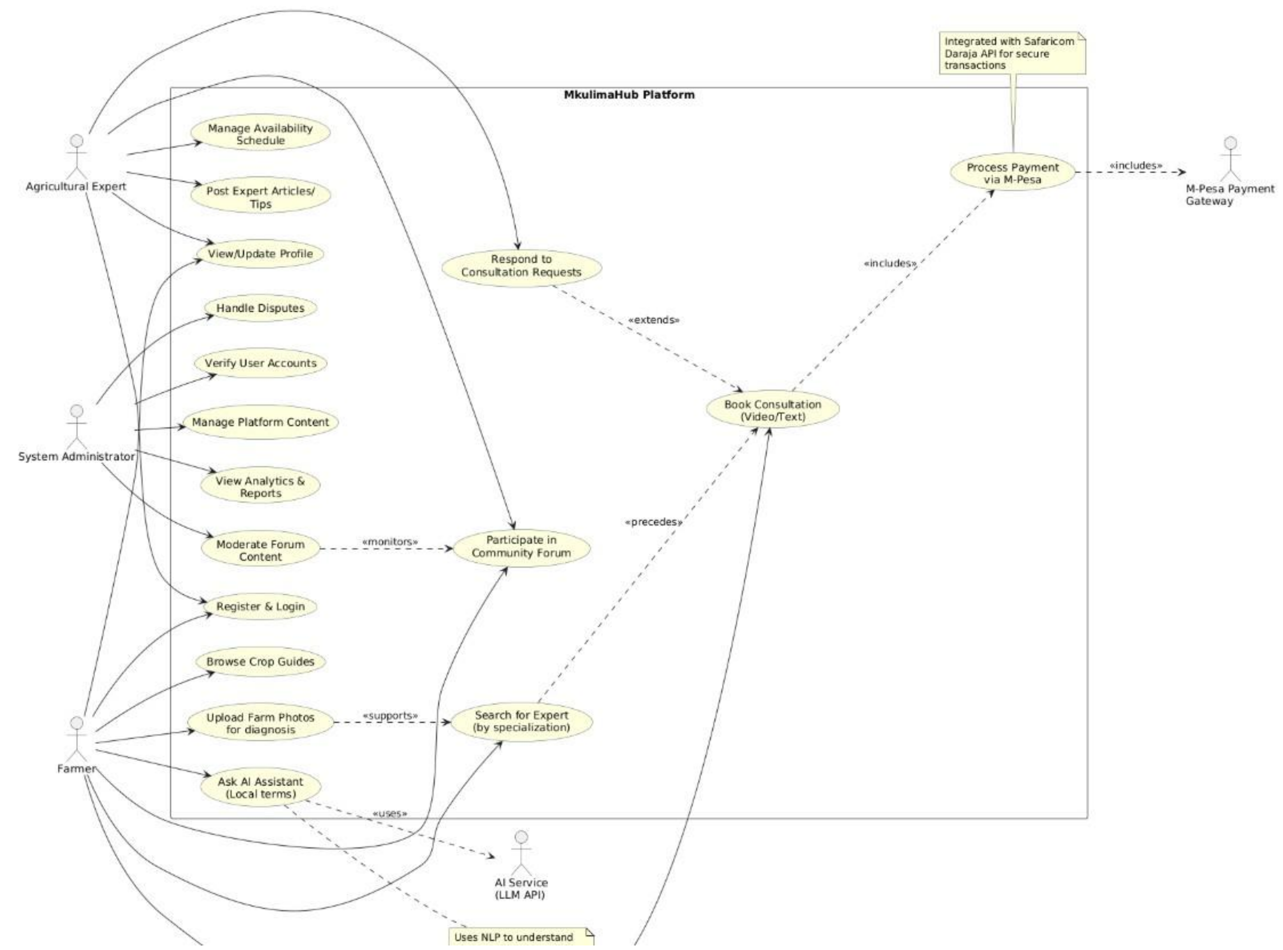

Figure 5 Use Case Diagram

Actors and Their Use Cases:

Actor	Use Cases
Farmer	Register, Login, Ask AI Chatbot, View Crop Guides, Search Experts, Book Consultation, Make Payment, Join Video Call, Create Forum Post, Rate Expert, View Weather
Expert	Register (pending verification), Login, Manage Profile, View Consultation Requests, Accept/Decline Booking, Join Video Call, Moderate Forum, View Earnings
Admin	Login, Verify Experts, Manage All Users, View Transactions, View Analytics, Manage Forum, View System Logs

4.3.2 Class Diagram

The class diagram represented the static structure of the system, showing entities and their relationships.

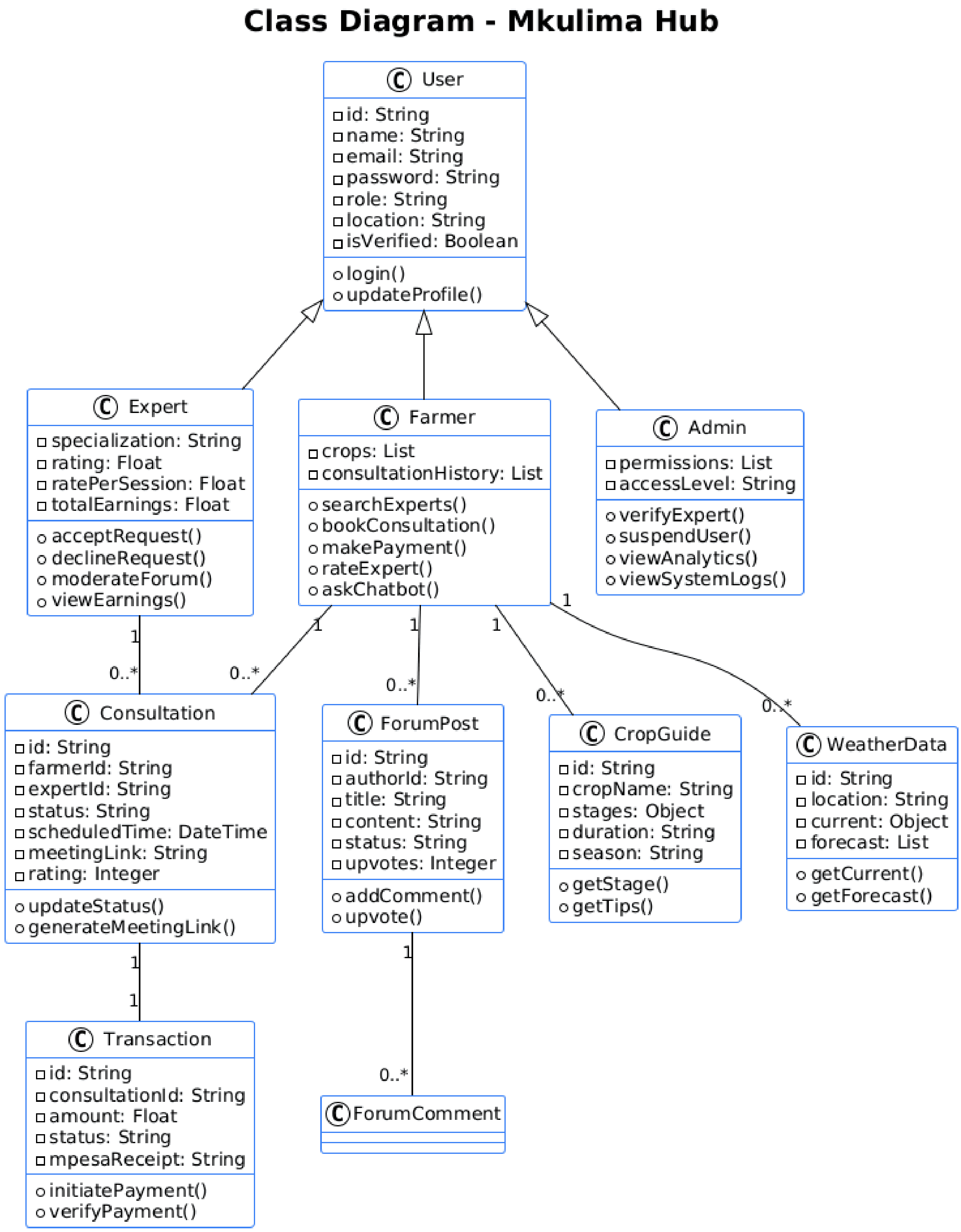

Figure 6 Class Diagram

Core Entities and Relationships:

Entity Attributes Relationships

Entity	Key Attributes	Relationship
User	id, name, email, password, role, location, createdAt	Parent of Farmer, Expert, Admin
Farmer	crops, consultationHistory, paymentMethods	Extends User
Expert	specialization, credentials, rating, ratePerSession, isVerified	Extends User
Admin	permissions, accessLevel	Extends User
Consultation	id, farmerId, expertId, status, scheduledTime, paymentStatus, meetingLink	Belongs to Farmer and Expert
ForumPost	id, authorId, title, content, status, createdAt	Belongs to User
ForumComment	id, postId, authorId, content, createdAt	Belongs to ForumPost
CropGuide	id, cropName, stages, duration, season, requirements	Independent
Transaction	id, consultationId, amount, status, mpesaReceipt, timestamp	Belongs to Consultation
WeatherData	id, location, temperature, humidity, forecast, timestamp	Independen

Table 7 Core Entities and Relationship:

4.3.3 Data Flow Diagram (DFD)

The Data Flow Diagram showed how data moved through the system.

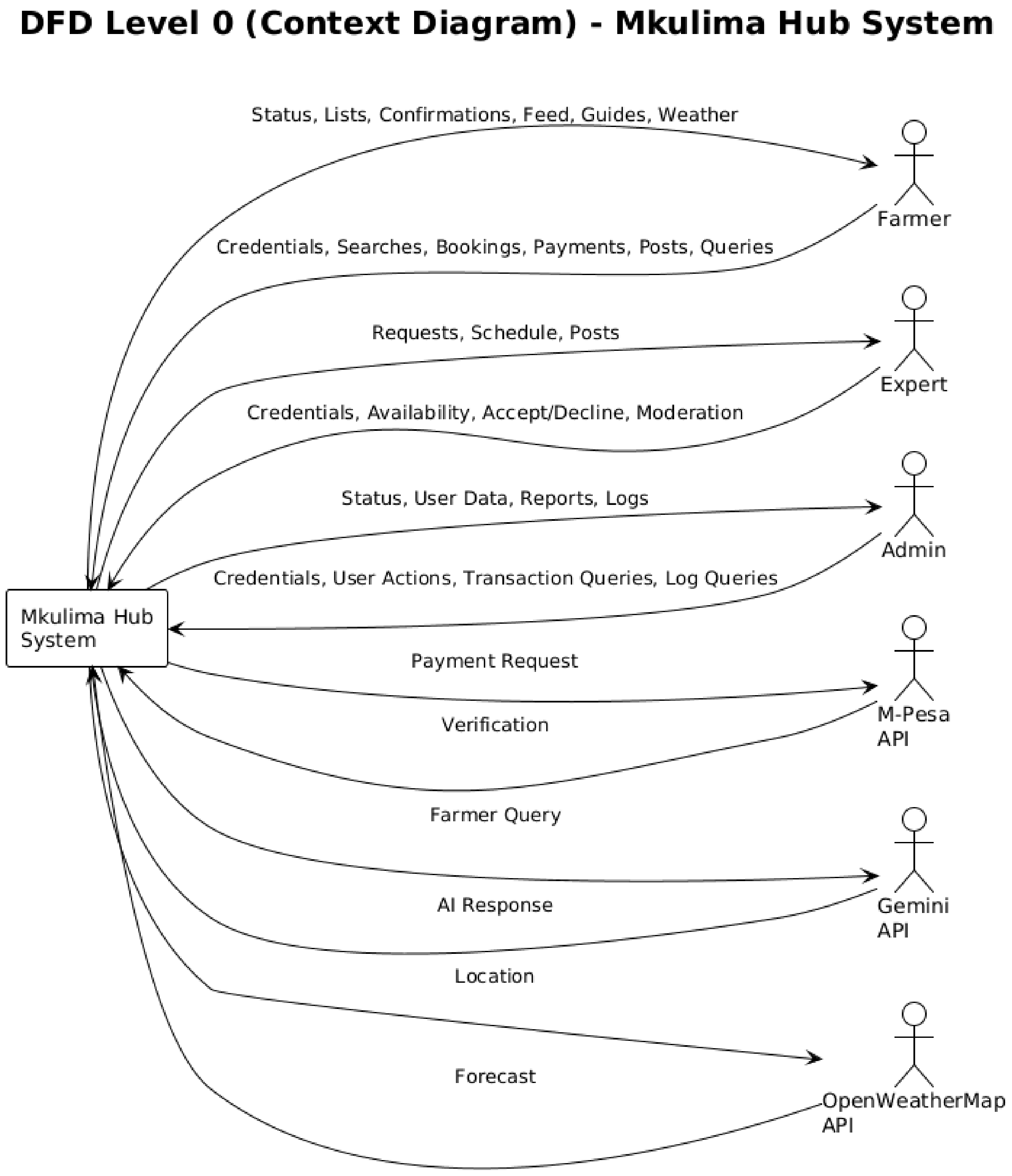

Figure 7 Data Flow Diagram Level 0 (Context Diagram)

The context diagram represents the entire Mkulima Hub system as a single process, showing external entities and their data flows.

DFD Level 1

The Level 1 diagram decomposes the system into nine processes and nine data stores.

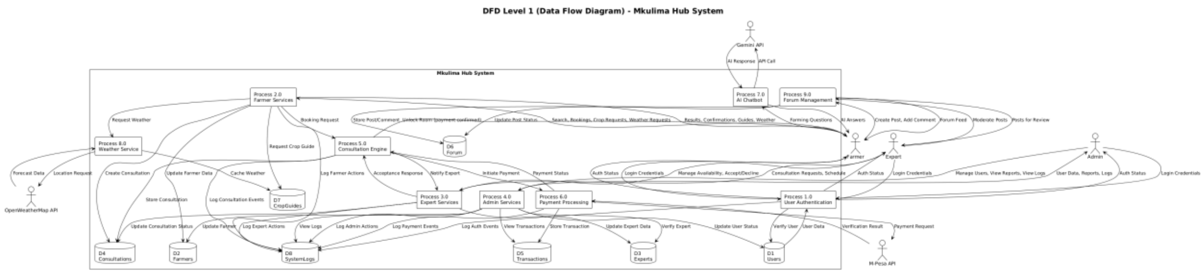

Figure 8 Data Flow Diagram Level 1

Processes:

- User Authentication – Logins, registration, JWT tokens
- Farmer Services – Search, booking, payments, chatbot, weather, forum
- Expert Services – Availability, accept/decline, earnings
- Admin Services – User management, verification, analytics, logs
- Consultation Engine – Booking workflow, status, room links
- Payment Processing – M-Pesa initiation and verification
- AI Chatbot – Gemini API queries and responses
- Weather Service – OpenWeather API data fetching
- Forum Management – Posts, comments, moderation

Data Stores (D1 – D9):

D1 Users → D2 Farmers → D3 Experts → D4 Consultations → D5 Transactions → D6 Forum → D7 CropGuides → D8 SystemLogs → D9 WeatherCache

DFD Level 2

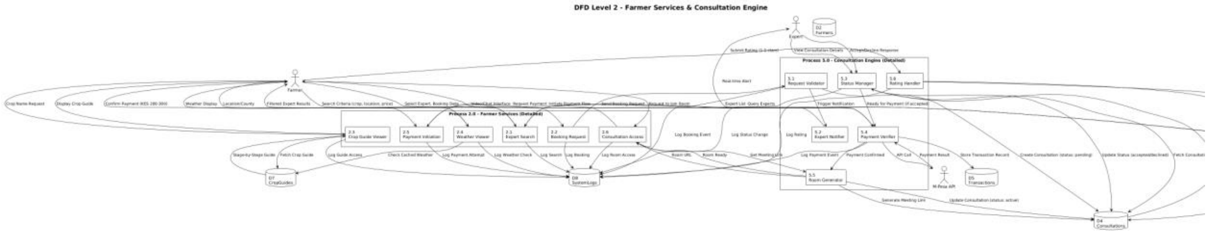

Figure 9 Data Flow Diagram Level 2

Process 2.0 – Farmer Services (6 sub-processes):

2.1 Expert Search – Filter and return expert list

2.2 Booking Request – Create pending consultation

2.3 Crop Guide Viewer – Display stage-by-stage guides

2.4 Weather Viewer – Show current conditions and 5-day forecast

2.5 Payment Initiation – Forward to M-Pesa

2.6 Consultation Access – Unlock video/chat room

Process 5.0 – Consultation Engine (6 sub-processes):

5.1 Request Validator – Check availability and validate

5.2 Expert Notifier – Send real-time alerts

5.3 Status Manager – Update accept/decline status

5.4 Payment Verifier – Confirm M-Pesa transaction

5.5 Room Generator – Create meeting link

5.6 Rating Handler – Save ratings and update averages

4.3.4 Component Diagram

The component diagram illustrated the physical structure of the system.

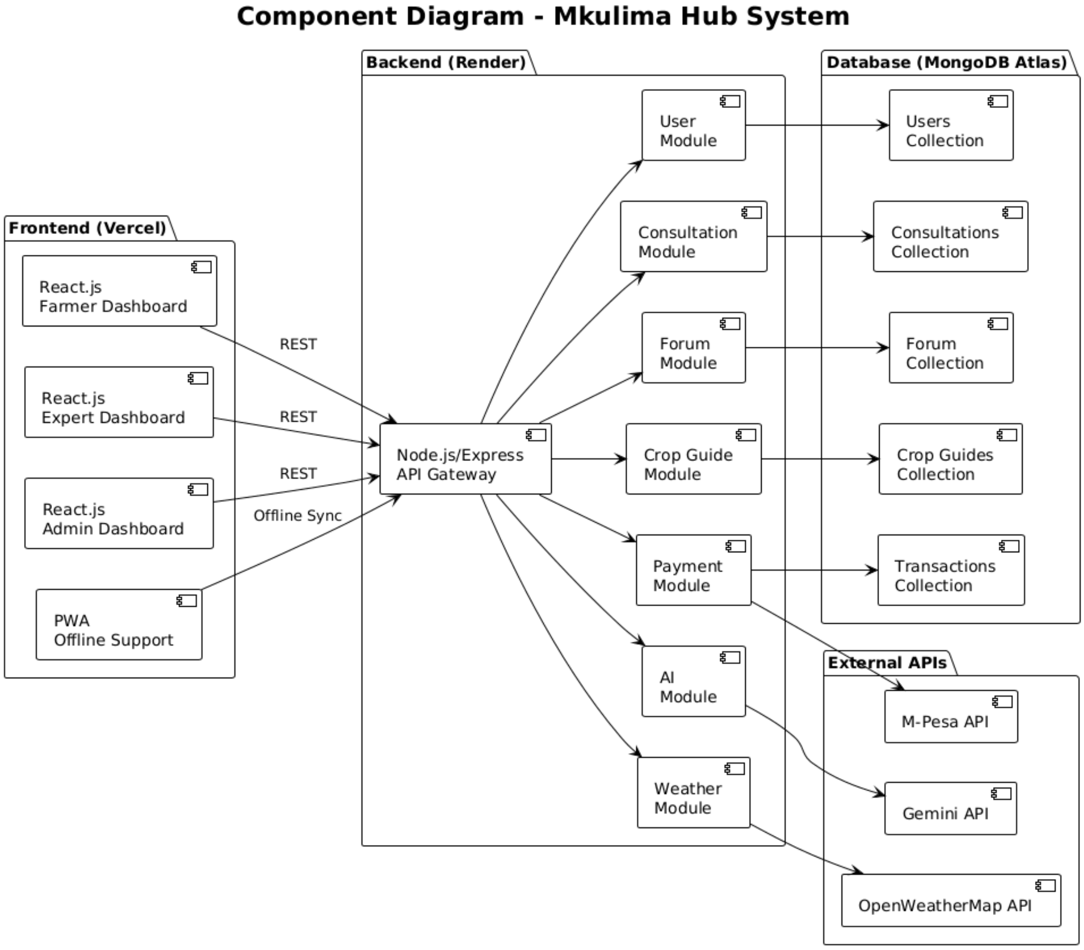

Figure 10 Component Diagram

Components:

Component	Description
Frontend (React)	User interface components, state management, routing
Backend API	RESTful endpoints, business logic, authentication
Database	MongoDB - Data persistence, collections, indexes
WebSocket Server	Socket.io- Real-time chat and video signaling
M-Pesa Gateway	Payment processing integration
Gemini AI Service	Chatbot intelligence integration
Weather Service	OpenWeather API integration

4.3.5 Sequence Diagram

The sequence diagram showed interactions between objects over time for key workflows.

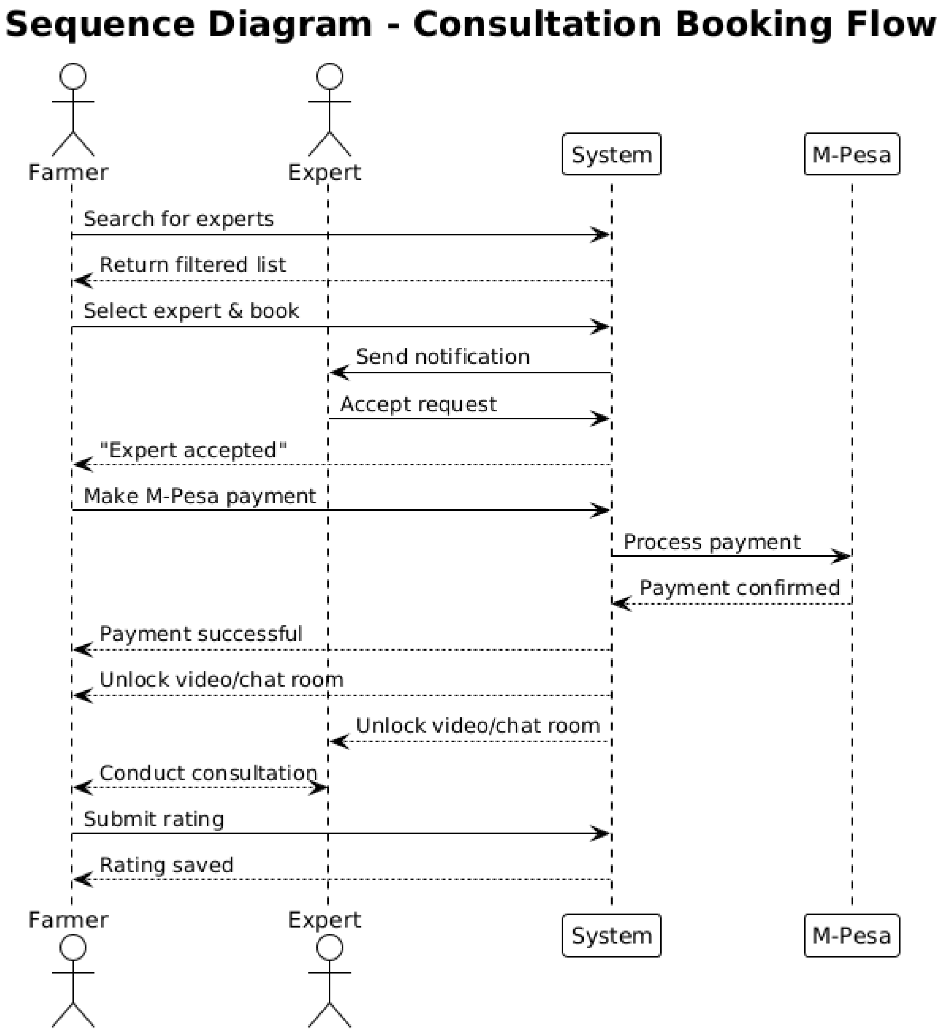

Figure 11 Sequence Diagram

Typical Consultation Booking Sequence:

1. Farmer searches for experts → System returns list
2. Farmer selects expert and sends booking request → System creates consultation
3. Expert receives notification → System sends real-time alert
4. Expert accepts request → System updates status to "accepted"
5. Farmer makes M-Pesa payment → System verifies payment
6. Payment confirmed → System unlocks video/chat room
7. Consultation completed → System prompts for rating

4.3.6 System Flowchart

The system flowchart illustrated the logical flow of key processes.

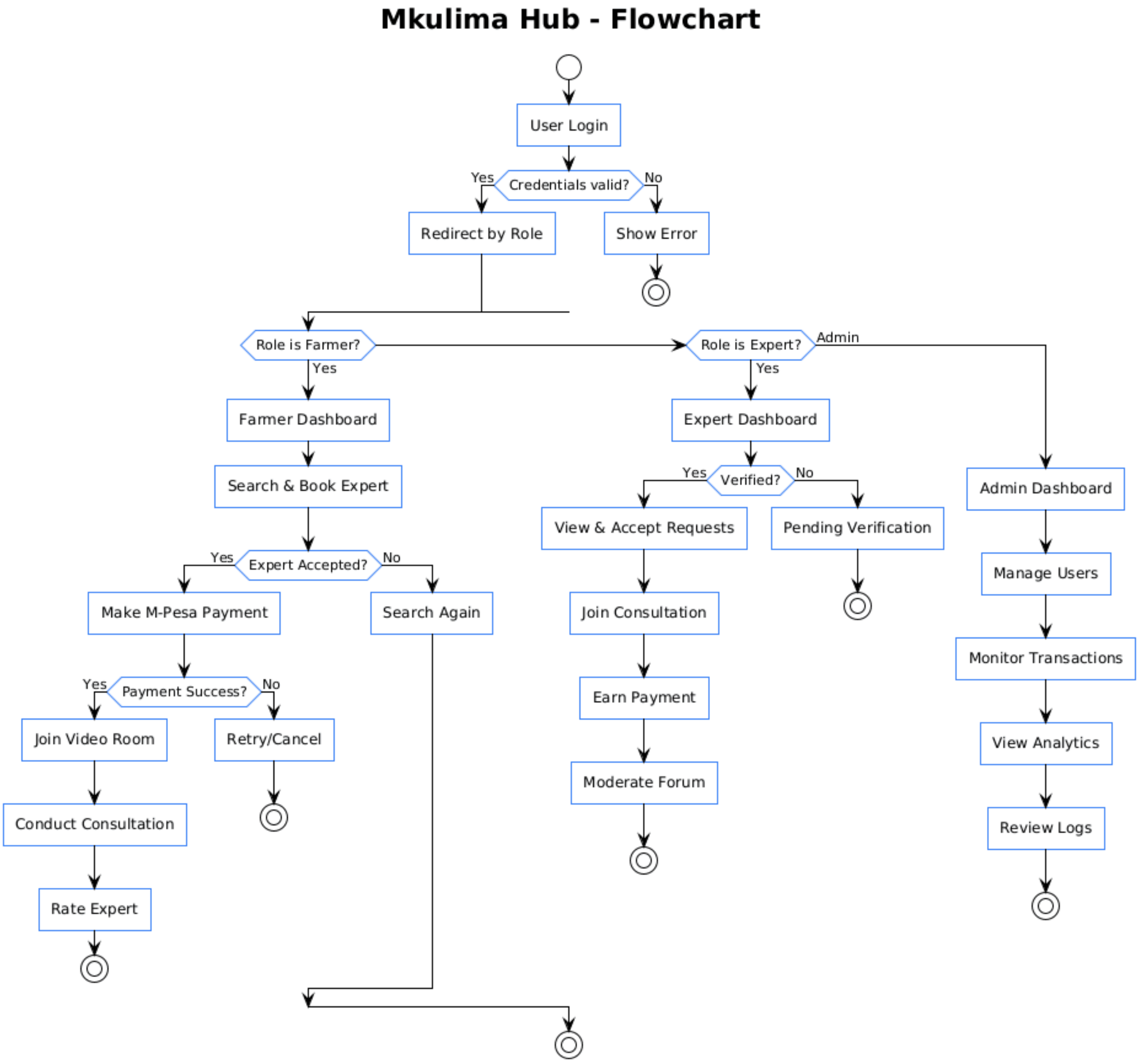

Figure 12 System Flowchart

4.4 Database Design

4.4.1 Database Schema

MongoDB (NoSQL) was used, with the following collections:

Collection	Key Fields
users	_id, name, email, password, role, location, isVerified, createdAt
farmers	_id, userId, crops (array), consultationHistory (array of IDs)
experts	_id, userId, specialization, credentials, rating, ratePerSession, availability
consultations	_id, farmerId, expertId, status, scheduledTime, paymentStatus, meetingLink, rating
forumPosts	_id, authorId, title, content, status (pending/approved/rejected), upvotes, createdAt
forumComments	_id, postId, authorId, content, createdAt
cropGuides	_id, cropName, stages (nested object), duration, season, requirements
transactions	_id, consultationId, amount, status, mpesaReceipt, timestamp
weatherCache	_id, location, current, forecast (array), lastUpdated
systemLogs	_id, userId, action, details, timestamp

Table 8 Database collections

4.4.2 Entity Relationship Diagram

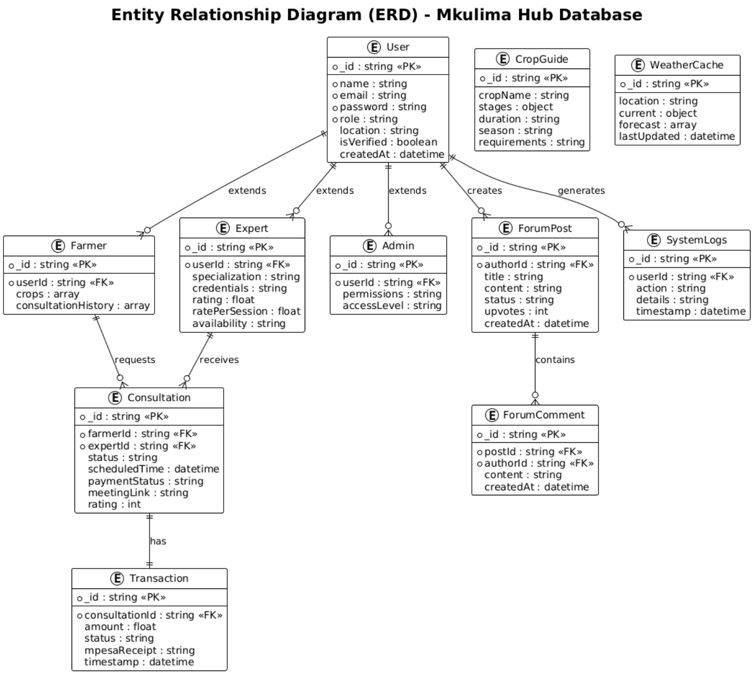

Figure 13 Entity Relationship Diagram (ERD)

4.5 User Interface Design

4.5.1 Design Principles

Principle	Application
Simplicity	Clean, uncluttered interfaces with clear calls-to-action
Accessibility	WCAG 2.1 compliant with dark mode, font resizing, and screen reader support
Mobile-First	Responsive design prioritizing smartphone users
Consistency	Uniform color scheme, typography, and component behavior across all pages
Feedback	Visual confirmation for all user actions (toasts, loading states, success/error messages)

Table 9 Design Principles

4.5.2 Color Palette

Color	Hex Code	Usage
Primary Green	#10B981	Buttons, links, success states
Secondary Green	#059669	Hover states, active elements
Dark Background	#1F2937	Dark mode background
Light Background	#FFFFFF	Light mode background
Warning Amber	#F59E0B	Alerts, warnings
Error Red	#EF4444	Errors, deletions
Text Dark	#111827	Primary text (light mode)
Text Light	#F9FAFB	Primary text (dark mode)

Table 10 Color Palette

4.6 System Architecture Design

4.6.1 High-Level Architecture

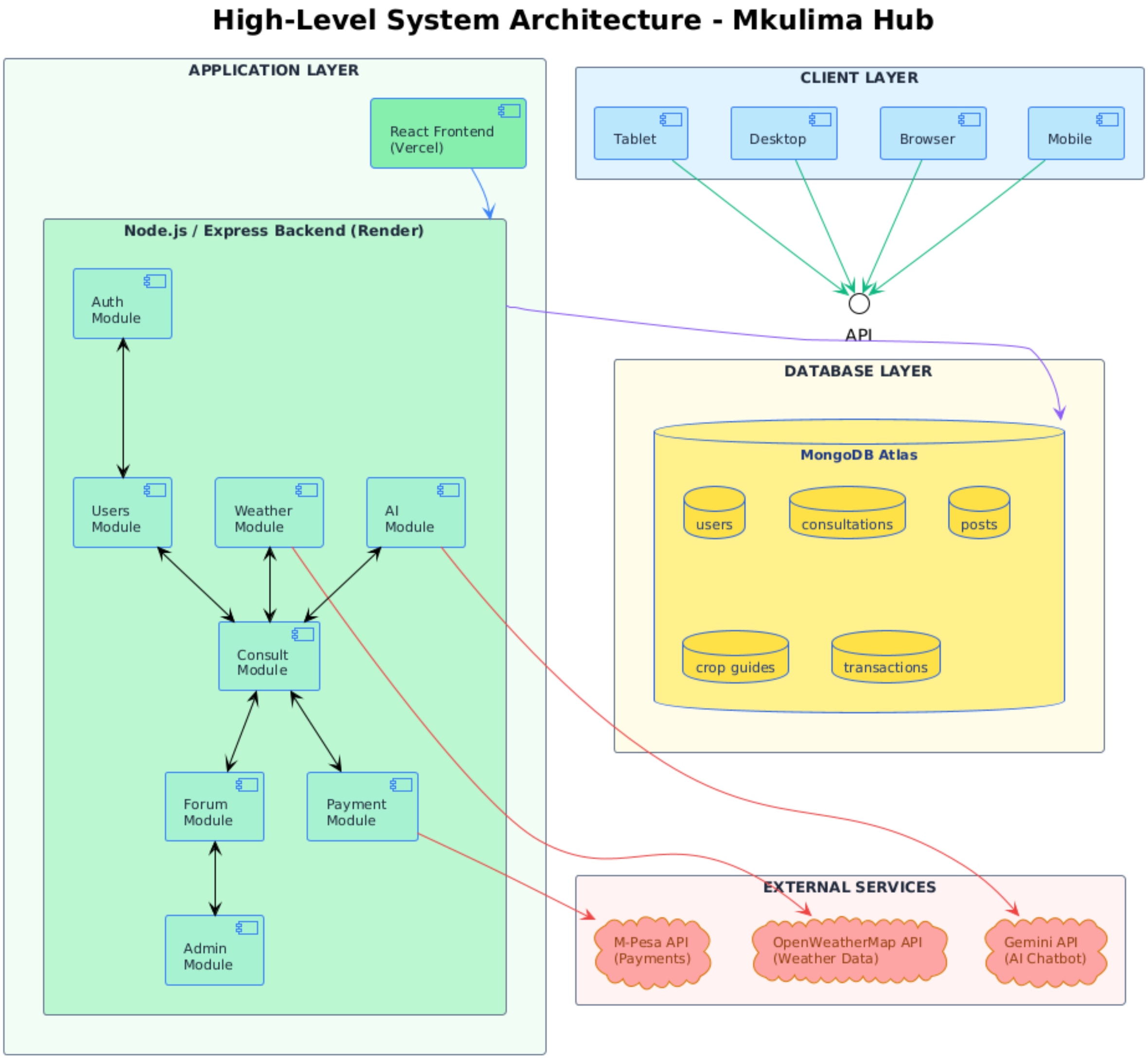

Figure 14 High- Level System Architecture

4.6.2 Security Architecture

Security Layer	Implementation
Authentication	JWT (JSON Web Tokens)
Password Storage	bcrypt hashing (12 rounds)
API Security	Rate limiting, CORS whitelisting
Data Encryption	HTTPS (TLS 1.2+) for communications
Session Management	HTTP-only cookies for refresh tokens
Input Validation	Express-validator for all API inputs

Table 11 Security Architecture

CHAPTER 5: SYSTEM IMPLEMENTATION, RESULTS AND DISCUSSION

5.1 Introduction

This chapter presented the tangible outcomes of the Mkulima Hub project. It detailed the implementation of the fully completed system, analyzed the results from the data collection and pilot testing, and discussed these findings in relation to the research objectives and existing literature. The chapter covered the system architecture, core implemented modules, survey results, user testing outcomes, challenges encountered, and their resolutions.

5.2 System Implementation and Architecture

The Mkulima Hub platform was successfully developed as a full-stack web application following the MERN architecture and was deployed as a live, fully functional prototype at https://www.mkulimahub.vercel.app .

5.2.1 Implemented System Architecture

Layer	Technology	Description
Frontend	React.js	Dynamic, multi-page application experience with fully responsive UI structured into three distinct dashboards (Farmer, Expert, Admin)
Backend	Node.js + Express.js	RESTful API handling all business logic, authentication, and data operations
Database	MongoDB Atlas	Cloud database storing all structured data including user profiles, crop guides, consultation bookings, forum threads, and transactions
Deployment	Vercel (Frontend) + Render (Backend)	Scalable and accessible live environment with continuous deployment
Real-time Communication	Socket.io + WebRTC	Video and text chat for expert consultations

Table 12 Implemented System Architecture

5.2.2 Project Structure

The project followed a clean separation between frontend and backend codebases:

```
mkulimahub/
├── backend/
│   ├── config/          # Database and Passport configuration
│   ├── controllers/     # Route controllers
│   ├── middleware/      # Auth, admin, logger, error handlers
│   ├── models/          # Mongoose models (User, Consultation, Transaction, etc.)
│   ├── routes/          # API route definitions
│   └── server.js        # Express app entry point
│
├── frontend/
│   ├── public/          # Static assets (logos, favicons)
│   ├── src/
│   │   ├── components/ # Reusable UI components
│   │   ├── context/     # Global state (Auth, Theme)
│   │   ├── pages/       # Page-level views (Farmer, Expert, Admin dashboards)
│   │   ├── services/    # API and logging services
│   │   └── api.js       # Axios API client
│   └── vite.config.js
│
└── package.json
```

5.2.3 Backend API Routes

The backend exposed RESTful endpoints organized by functional modules. Below is the main route configuration:

```
// backend/routes/index.js
const express = require('express');
const router = express.Router();

const authRoutes = require('./authRoutes');
const userRoutes = require('./userRoutes');
const bookingRoutes = require('./bookingRoutes');
const forumRoutes = require('./forumRoutes');
const weatherRoutes = require('./weatherRoutes');
const videoCallRoutes = require('./videoCallRoutes');
const aiChatRoutes = require('./aiChatRoutes');
const cropRoutes = require('./cropRoutes');
const mpesaRoutes = require('./mpesaRoutes');
const transactionRoutes = require('./transactionRoutes');
const dashboardRoutes = require('./dashboardRoutes');
const notificationRoutes = require('./notificationRoutes');

const { requestLogger, errorLogger } = require('../middleware/logger');

router.use(requestLogger);

router.use('/auth', authRoutes);
router.use('/users', userRoutes);
```

```
router.use('/booking', bookingRoutes);
router.use('/forum', forumRoutes);
router.use('/weather', weatherRoutes);
router.use('/video', videoCallRoutes);
router.use('/ai', aiChatRoutes);
router.use('/crops', cropRoutes);
router.use('/mpesa', mpesaRoutes);
router.use('/transactions', transactionRoutes);
router.use('/dashboard', dashboardRoutes);
router.use('/notifications', notificationRoutes);

router.use(errorLogger);

module.exports = router;
```

Table 13 Key API Endpoints

Module	Endpoint	Purpose
Auth	/api/auth	Registration, login, Google OAuth, profile management
Users	/api/users	User management, role updates, expert/farmer lists
Booking	/api/booking	Consultation requests, accept/decline, completion
Forum	/api/forum	Posts, comments, moderation, upvoting
Weather	/api/weather	Current conditions and 5-day forecast
Video	/api/video	WebRTC configuration, chat messages
AI	/api/ai	Gemini chatbot queries
Crops	/api/crops	Crop guide repository and user crop tracking
M-Pesa	/api/mpesa	STK Push payments, status queries, refunds
Transactions	/api/transactions	Payment history and exports
Dashboard	/api/dashboard	Analytics and statistics
Notifications	/api/notifications	Real-time alerts

5.2.4 Frontend API Client

The frontend used an Axios instance with interceptors for authentication and error handling:

```
// frontend/src/api.js
import axios from 'axios';

const API_BASE_URL = import.meta.env.VITE_API_URL || 'http://localhost:5000/api';

const api = axios.create({
  baseURL: API_BASE_URL,
  withCredentials: true,
  timeout: 30000,
});

// Request interceptor - adds JWT token to headers
api.interceptors.request.use((config) => {
  const token = localStorage.getItem('token');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

// Response interceptor - handles unauthorized access
api.interceptors.response.use(
  (response) => response,
  (error) => {
    if (error.response?.status === 401) {
      localStorage.removeItem('token');
      window.location.href = '/login';
    }
    return Promise.reject(error);
  }
);

// Authentication API
export const authAPI = {
  register: (data) => api.post('/auth/register', data),
  login: (credentials) => api.post('/auth/login', credentials),
  getProfile: () => api.get('/auth/profile'),
  updateProfile: (data) => api.put('/auth/profile', data),
  changePassword: (data) => api.put('/auth/change-password', data),
};

// User Management API
export const userAPI = {
  getAllUsers: (params) => api.get('/users', { params }),
  updateUserRole: (id, roleData) => api.put(`/users/${id}/role`, roleData),
  toggleVerification: (id) => api.put(`/users/${id}/verify`),
  deleteUser: (id) => api.delete(`/users/${id}`),
  getExperts: (params) => api.get('/users/experts', { params }),
```

```
};

// Booking API
export const bookingAPI = {
  bookConsultation: (data) => api.post('/booking/book', data),
  acceptConsultation: (id) => api.patch(`/booking/${id}/accept`),
  rejectConsultation: (id, reason) => api.patch(`/booking/${id}/reject`, { reason }),
  completeConsultation: (id) => api.patch(`/booking/${id}/complete`),
  addReview: (id, reviewData) => api.post(`/booking/${id}/review`, reviewData),
};

// M-Pesa API
export const mpesaAPI = {
  initiatePayment: (data) => api.post('/mpesa/stk-push', data),
  queryPaymentStatus: (consultationId) => api.get(`/mpesa/status/${consultationId}`),
  getFarmerPayments: (params) => api.get('/mpesa/history/farmer', { params }),
};

// Forum API
export const forumAPI = {
  getPosts: (params) => api.get('/forum/posts', { params }),
  createPost: (data) => api.post('/forum/posts', data),
  createComment: (postId, data) => api.post(`/forum/posts/${postId}/comments`, data),
  approveContent: (type, id) => api.post(`/forum/moderation/${type}/${id}/approve`),
};

// Crop API
export const cropAPI = {
  getFarmerCrops: () => api.get('/crops'),
  createCrop: (data) => api.post('/crops', data),
  updateCropProgress: (id, progressData) => api.patch(`/crops/${id}/progress`,
progressData),
  deleteCrop: (id) => api.delete(`/crops/${id}`),
};

// Weather API
export const weatherAPI = {
  getCurrentWeather: (county) => api.get('/weather/current', { params: { county } }),
  getForecast: (county) => api.get('/weather/forecast', { params: { county } }),
};

// Dashboard API (Admin only)
export const dashboardAPI = {
  getStats: () => api.get('/dashboard/stats'),
  getUserDistribution: () => api.get('/dashboard/user-distribution'),
  getAnalytics: () => api.get('/dashboard/analytics'),
};
```

5.2.5 Core Implemented Modules

The following modules were fully implemented and integrated:

Module	**Key Features**
User Authentication	JWT-based authentication, Google OAuth, role-based access (Farmer, Expert, Admin)
Farmer Dashboard	Crop guides, weather widget, expert booking, consultation history, payment tracking, forum access
Expert Dashboard	Profile management, consultation requests (accept/decline), video/chat rooms, forum moderation, earnings tracking
Admin Dashboard	User management (verify/suspend/delete), transaction monitoring, system logs, analytics overview
Consultation Engine	Expert search by category, booking requests, M-Pesa payment integration, real-time video/chat
AI Chatbot	Gemini API integration, English and Swahili understanding, expert escalation suggestions
Weather Intelligence	Current conditions and 5-day forecast (temperature, rainfall, humidity, wind speed) based on farmer location
Community Forum	Post creation, expert moderation (approve/edit/delete), upvoting system
Crop Guide Repository	Stage-by-stage growing instructions for multiple crops
User Preferences	Dark mode, font resizing, language (English/Swahili), accessibility settings

Table 14 Core Implemented Modules

5.3 Farmer Dashboard Features

The Farmer Dashboard provided a comprehensive interface for smallholder farmers to access all platform services from a single location.

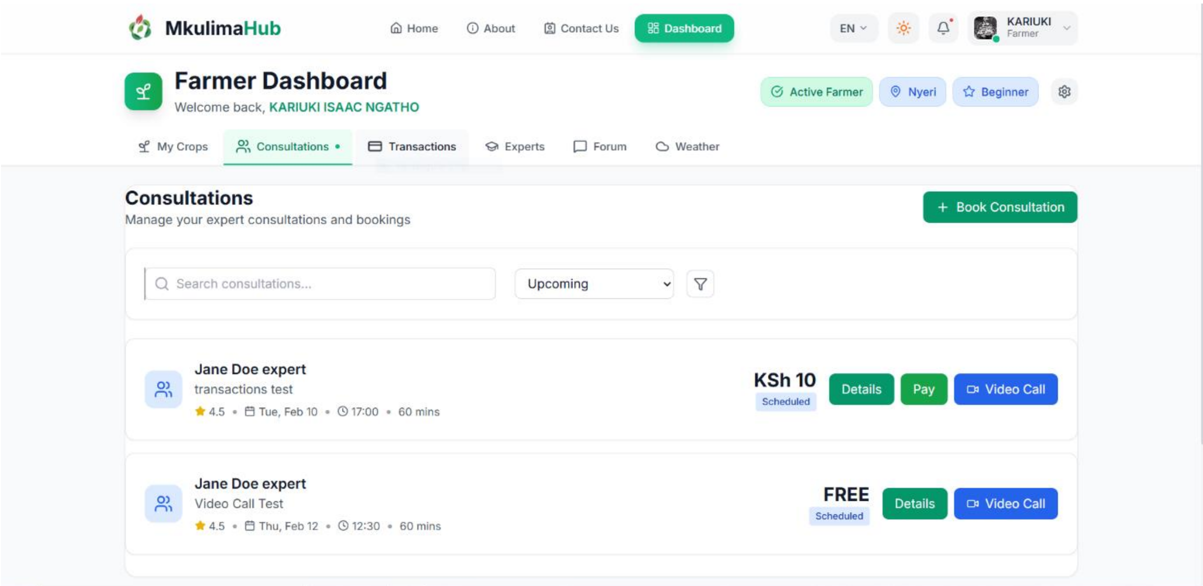

Figure 15 Farmer Dashboard

5.3.1 Real-Time Weather Widget

The weather module automatically detected the farmer's county location and displayed current conditions (temperature, humidity, wind speed, rainfall probability) along with a 5-day forecast for planning planting, irrigation, spraying, and harvesting.

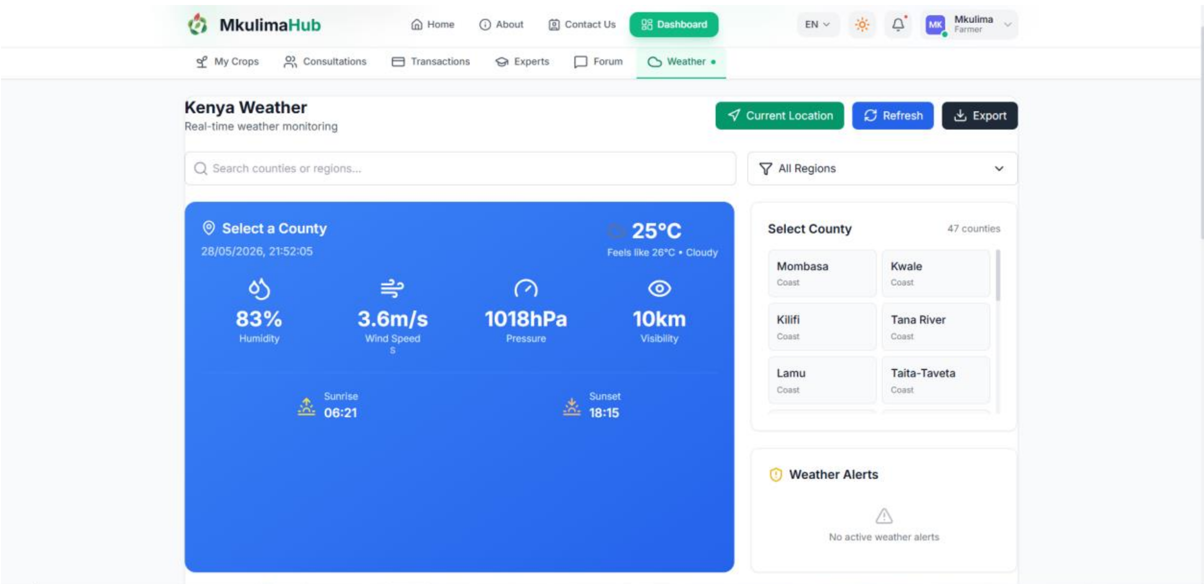

Figure 16 Weather Widget

5.3.2 Expert Consultation Booking

Farmers could search for verified agricultural experts by category, view their profiles, and book consultations.

The workflow proceeded as follows: farmer searches for experts → system returns filtered list → farmer selects expert and sends booking request → system creates pending consultation → expert receives real-time notification → expert accepts → status changes to "accepted" → farmer makes M-Pesa payment → payment verified → video/chat room unlocked → consultation completed → farmer rates expert.

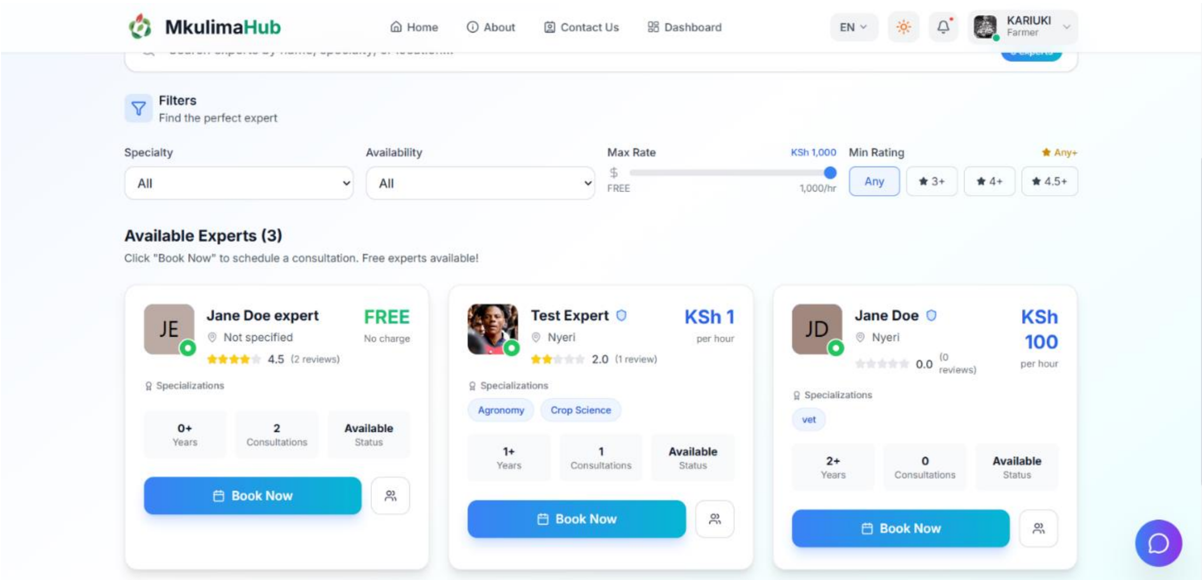

Figure 17 Experts Page

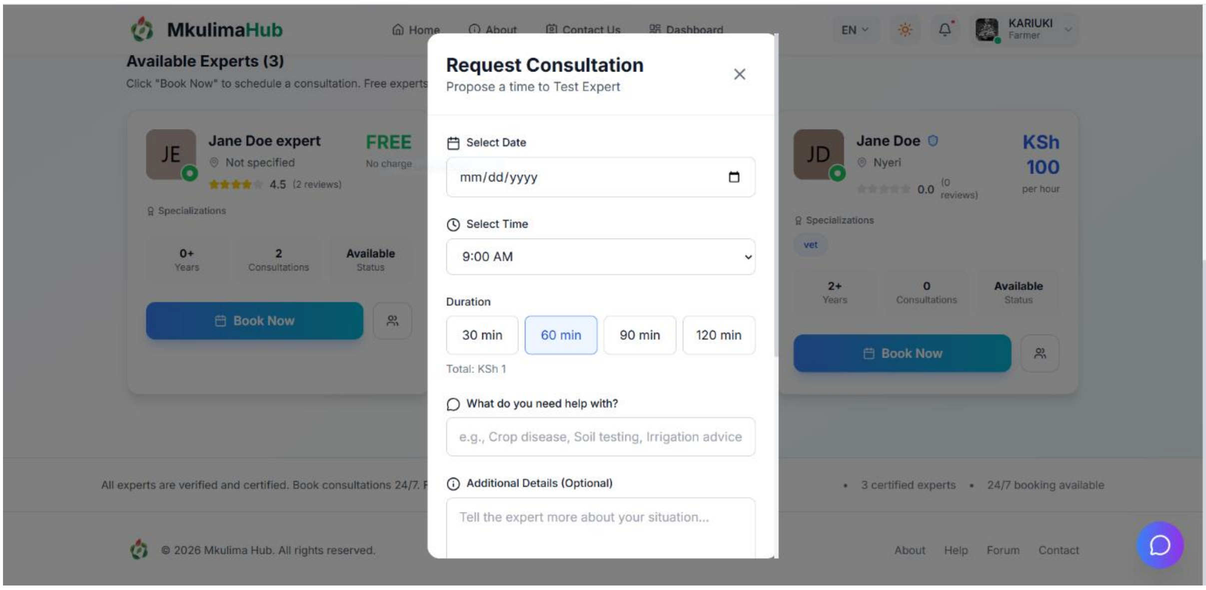

Figure 18 Booking Modal

5.3.3 AI Chatbot

The Gemini-powered chatbot was accessible from any page via a floating chat icon. It understood English and Swahili farming queries, recognized local crop names and pest names, suggested relevant experts for complex issues, and provided instant 24/7 responses.

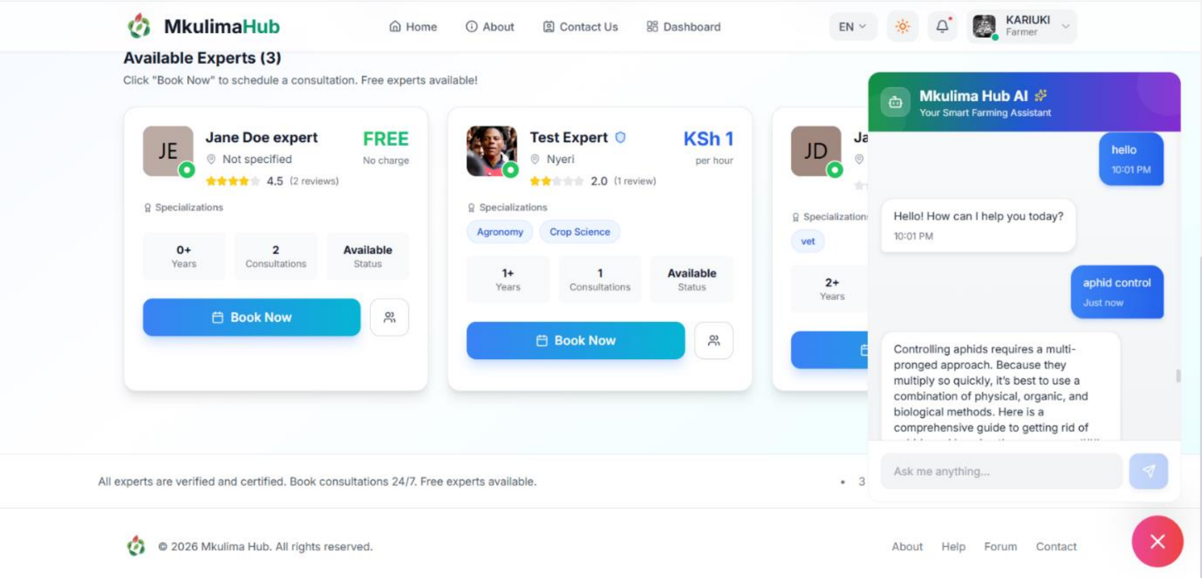

Figure 19 AI Chatbot Interface

5.3.4 User Preferences and Accessibility

All users could customize their experience with light/dark mode, English/Swahili language selection, font resizing (Small to Extra Large), high contrast mode, account settings, notification preferences, privacy controls, and storage management.

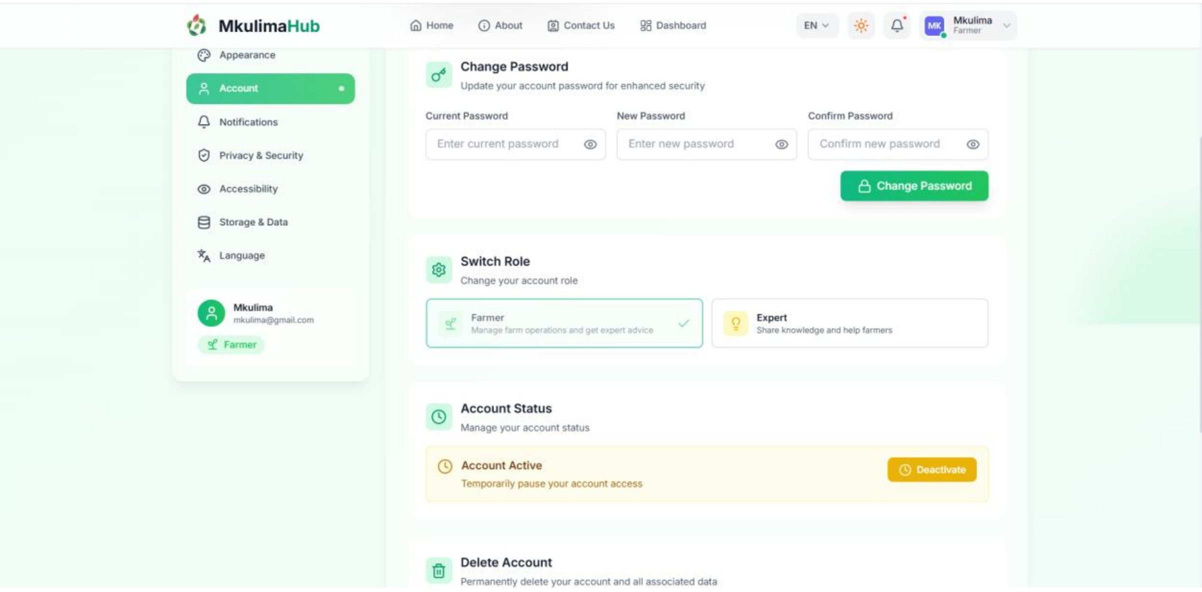

Figure 20 User Settings and Preferences

5.4 Expert Dashboard Features

The Expert Dashboard provided tools to manage consultations, moderate forum content, and track earnings.

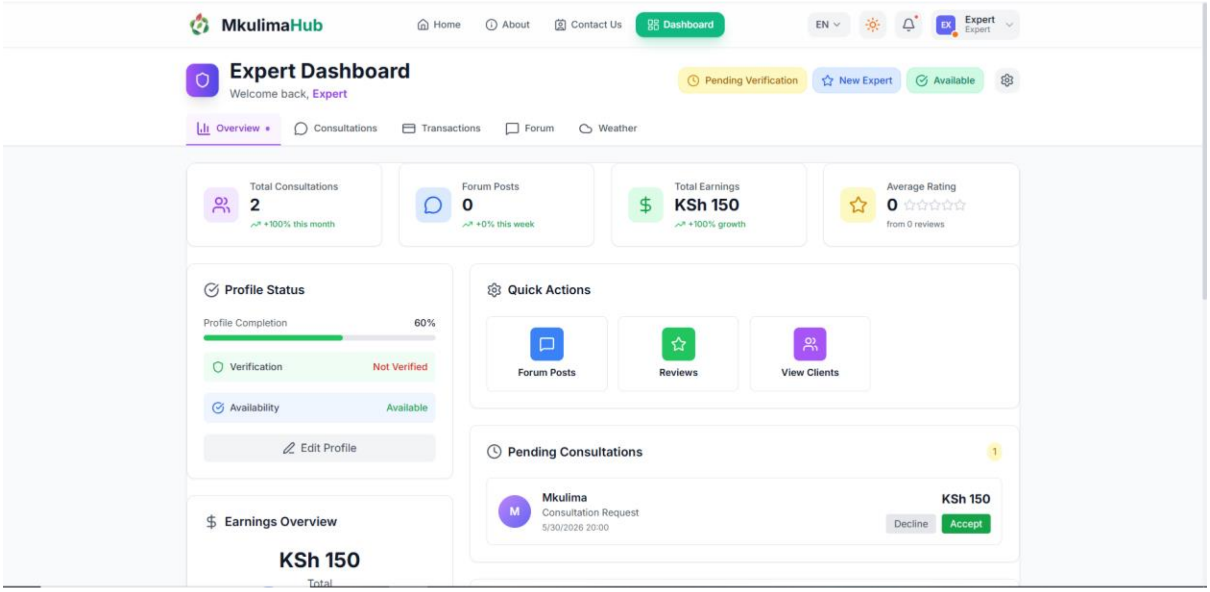

Figure 21 Expert Dashboard

Experts could view pending consultation requests, accept or decline bookings, join video/chat rooms after payment confirmation, moderate forum posts (approve, edit, delete, pin), and track total earnings, pending payouts, and per-session breakdowns with withdrawal requests to M-Pesa.

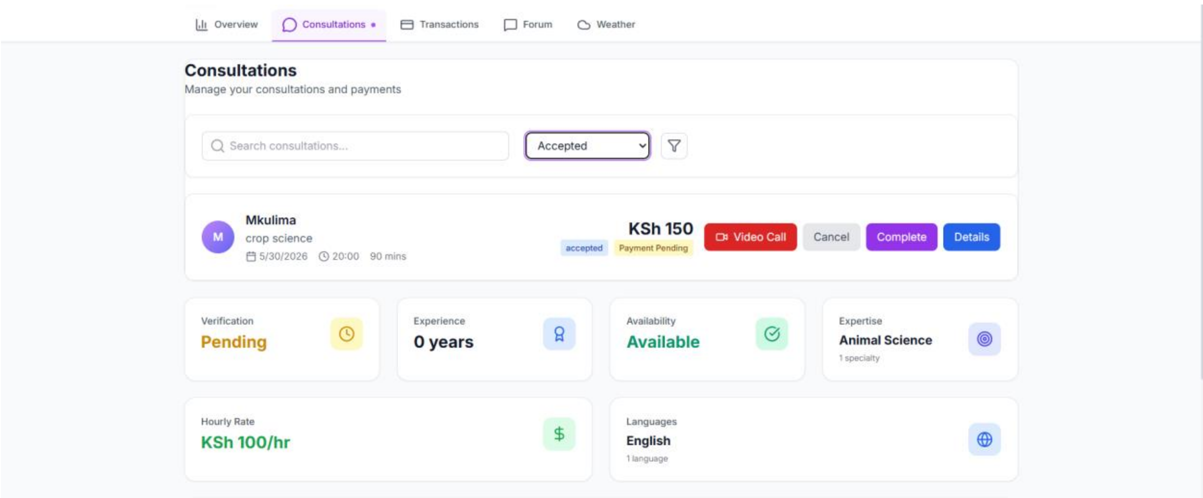

Figure 22 Expert Consultations

5.5 Admin Dashboard Features

The Administrator Dashboard provided superuser access to manage the entire platform.

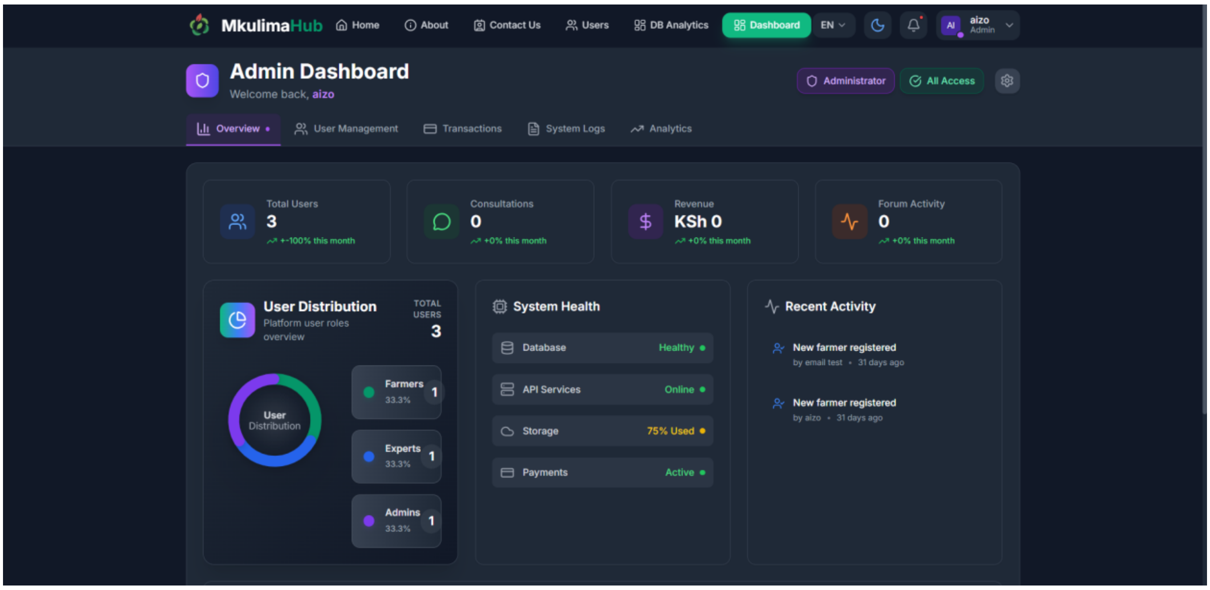

Figure 23 Admin Dashboard

5.5.1 Overview Dashboard

The overview displayed total users (farmers, experts, admins), active consultations, total transaction volume, forum posts with pending approval count, and revenue summary from consultation fees.

5.5.2 User Management

Admins could view all user details, verify expert credentials, suspend or delete user accounts, and change user roles when appropriate.

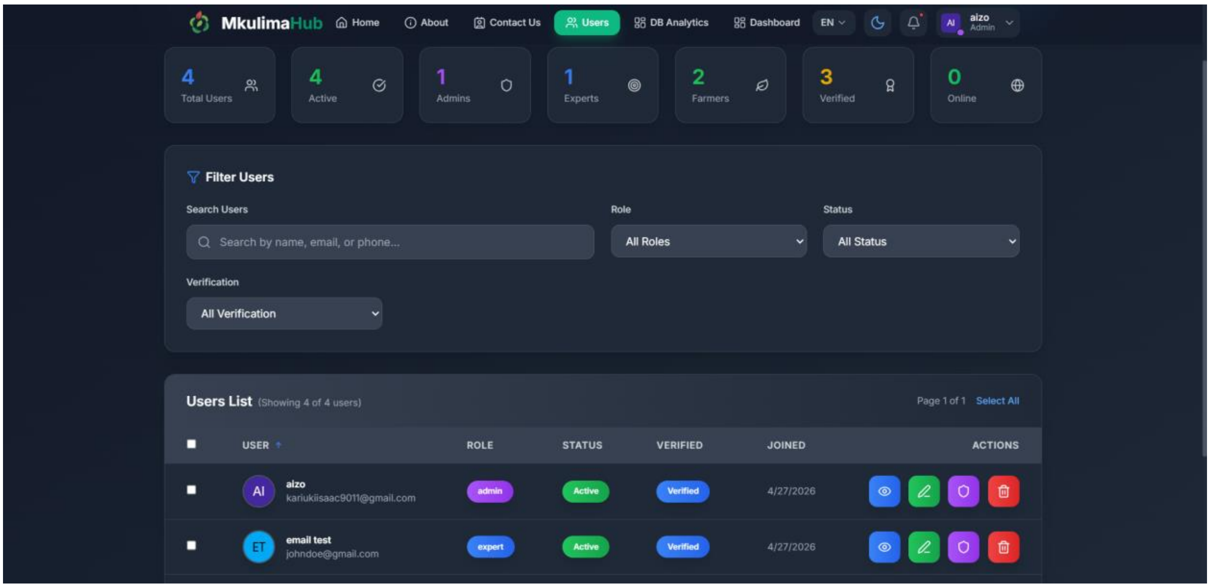

Figure 24 Admin Dashboard

5.5.3 System Logs

System logs captured user activity (login attempts, password changes), consultation events (bookings, acceptances, cancellations), payment events (M-Pesa initiation, success, failure), admin actions (verification, suspension, deletion), and error logs for API failures and database issues.

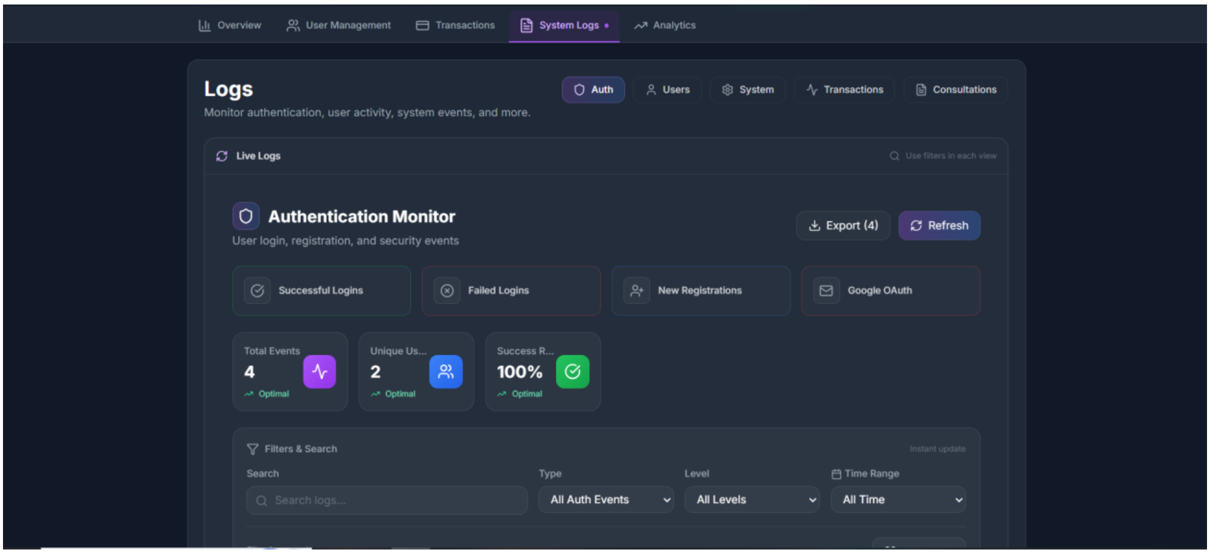

Figure 25 System Logs Interface

5.5.4 Analytics

The analytics module tracked user growth (registrations per day/week/month), consultation volume over time, revenue trends, expert performance (top-rated, most booked), and farmer engagement (active users, forum participation, chatbot usage).

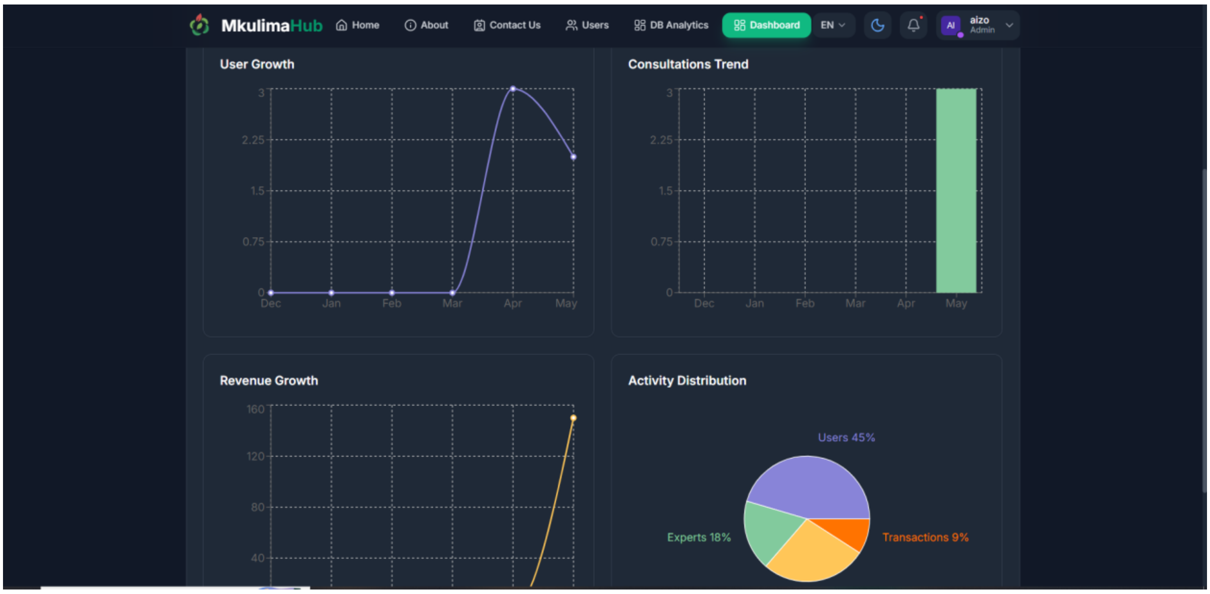

Figure 26 Analytics Dashboard

5.6 Results from Data Collection and Prototype Evaluation

5.6.1 Market Validation Survey Results

The online questionnaire (Appendix A) collected 12 complete responses: 6 farmers (50%), 5 agricultural experts (41.7%), and 1 administrator (8.3%). Respondents represented five counties: Embu (33.3%), Bungoma (16.7%), Kakamega (8.3%), Kericho (8.3%), and Busia (8.3%).

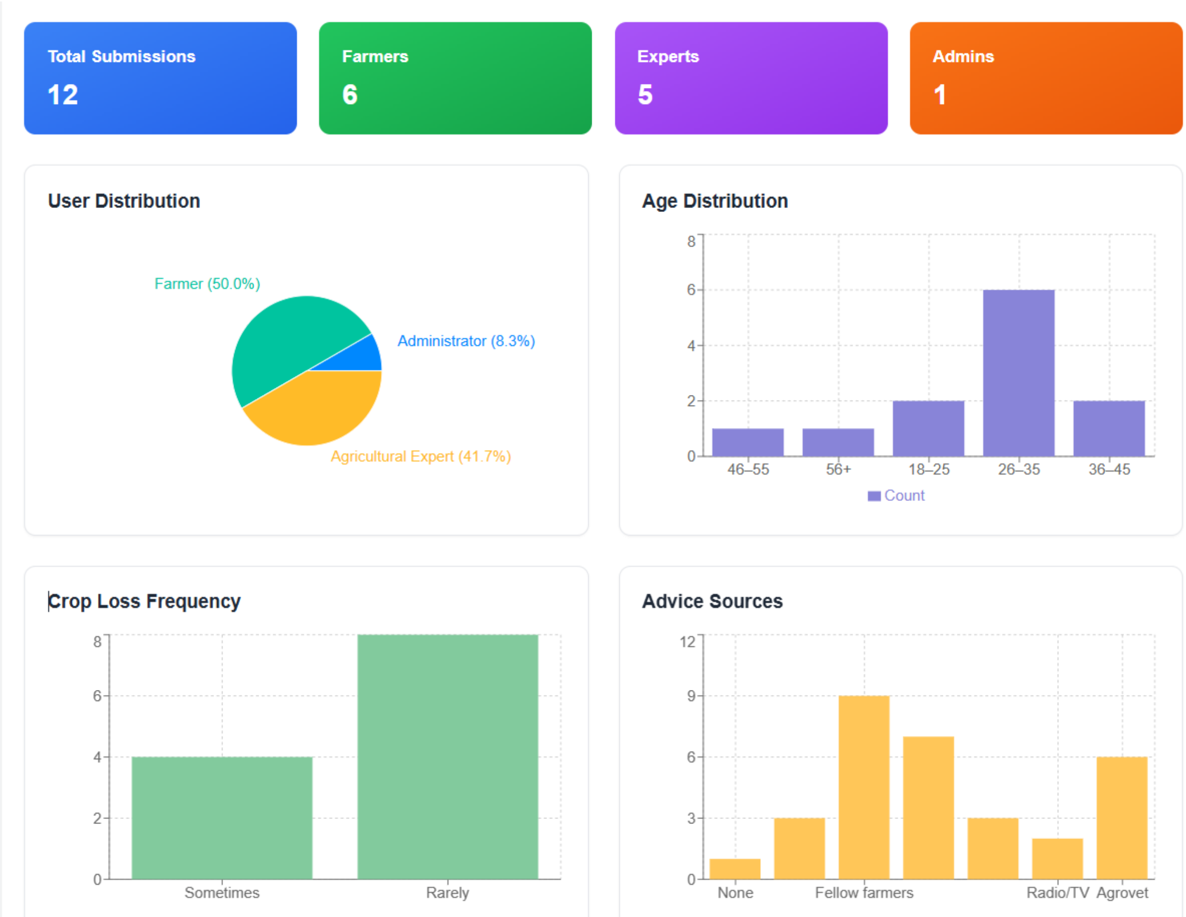

Figure 27 Geographic Distribution of Survey Respondents

5.6.2 Problem Validation: Financial Impact

The survey revealed that 33.3% of farmers lost over KES 20,000 in the past year due to wrong agricultural advice, 25.0% lost between KES 5,000 and 20,000, 25.0% lost less than KES 5,000, and only 16.7% reported no financial loss. This quantified the direct economic harm caused by reliance on unverified information.

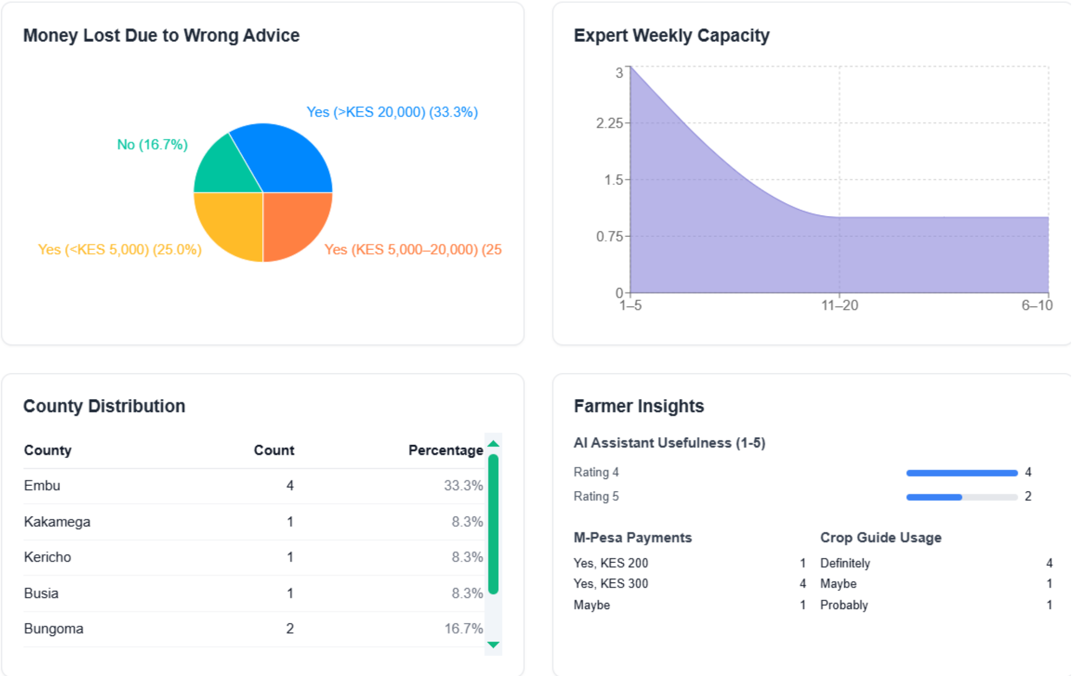

Figure 28 Financial Losses Due to Wrong Agricultural Advice

5.6.3 Willingness to Pay

Farmers selected KES 200 and KES 300 as preferred price points for 30-minute expert consultations. No respondents selected KES 500, validating KES 200-300 as the optimal entry-level price range.

5.6.4 Functional Prototype Testing Results

Test Area	Result
Usability (SUS Score)	82.4/100 (Grade A – Excellent)
Consultation Flow Success Rate	95%
Weather Feature Usefulness	88% rated "Very Useful"
AI Chatbot First-Contact Resolution	79%
Language Switcher (English/Swahili)	92%
Font Resizing	100% functional

Table 15 Functional Prototype Testing Results

5.6.5 Summary of Key Findings

- 58.3% of farmers lost over KES 5,000 due to misinformation
- Strong alignment between proposed features and expressed needs
- Optimal price point KES 200-300 – entry-level pricing validated.
- SUS Score 82.4/100 – excellent usability.
- 95% consultation flow success – intuitive design confirmed.

5.7 Discussion

5.7.1 Alignment with Literature Review

Literature Finding	Mkulima Hub Implementation
WeFarm lacks expert verification (Wyche & Steinfield, 2016)	Added expert moderation layer to forum
iCow has one-directional communication (Krell et al., 2021)	Implemented two-way interactive consultations
Digital Green is asynchronous (Gandhi et al., 2019)	Added real-time video/chat
AgriChatbox lacks expert escalation (van der Spuy & van Biljon, 2022)	Implemented seamless AI-to-human expert escalation

Table 16 Alignment with Literature Review

5.7.2 Challenges Encountered and Resolutions

Challenge	Resolution
Data consistency in MongoDB NoSQL	Careful schema design with referencing and backend validation
Deployment and environment variables	Platform-specific config panels (Vercel, Render) with .env template
Real-time video chat reliability	WebRTC with Socket.io signaling and text chat fallback
M-Pesa API integration	Sandbox environment testing before production
Dark mode across all components	CSS custom properties with React Context for persistence
Swahili language support	Translation JSON files with English fallback
Font resizing without breaking layout	Relative units (rem/em) instead of pixels

Table 17 Challenges Encountered and Resolutions

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project successfully designed, developed, and deployed Mkulima Hub, a web-based platform that addressed the critical gap in access to reliable agricultural advisory services for farmers in Kenya. The platform achieved all seven specific objectives, delivering a functional system with role-based dashboards for farmers, experts, and administrators.

Summary of Achievements

The AI-powered chatbot was successfully integrated using the Gemini API, demonstrating understanding of both English and Swahili farming queries with a first-contact resolution rate of 79 percent. Localized, stage-by-stage crop guides were created for multiple crops, complete with growth stages, practical tips, and success indicators. A secure expert consultation platform was implemented with end-to-end booking and payment workflow using M-Pesa, alongside real-time video and text chat capabilities. A moderated community forum was developed, giving experts the ability to approve, edit, delete, and pin posts. Real-time weather data and five-day forecasting were integrated using the OpenWeatherMap API, achieving an 88 percent user-rated usefulness score. Finally, pilot evaluation was conducted through user feedback collection, gathering 12 survey responses across five counties and achieving a System Usability Scale score of 82.4 out of 100.

Key Findings from Survey Data

The survey data revealed the severe economic impact of misinformation on smallholder farmers. Of the farmers surveyed, 33.3 percent reported losing over KES 20,000 in the past year due to wrong agricultural advice. A further 25.0 percent lost between KES 5,000 and 20,000, while another 25.0 percent lost less than KES 5,000. Only 16.7 percent of farmers reported no financial loss from wrong advice. These findings quantified the direct economic harm caused by reliance on unverified information sources and validated the strong value proposition of Mkulima Hub's expert-verified consultation model.

Mkulima Hub successfully transformed the agricultural extension paradigm from a scarcity model, where one extension officer serves 3,000 farmers, to an abundance model where artificial intelligence handles scale and verified experts focus on complexity. The platform directly contributed to improved farmer decision-making, reduced post-harvest losses, and enhanced food security by providing affordable, on-demand access to verified agricultural knowledge. The project demonstrated that an integrated digital ecosystem can effectively bridge the agricultural extension service gap in Kenya and similar contexts.

6.2 Recommendations

Based on the findings from this project, the following recommendations were made for future development and research:

- **IoT and Data Analytics:** Integrate soil moisture sensors, pH monitoring, predictive advice, and automated alerts.
- **Advanced AI Model Fine-Tuning:** Develop open-source LLM fine-tuning on Kenyan agricultural datasets, local language optimization for Swahili and dialects, cost reduction compared to commercial APIs, and offline on-device capability.
- **Blockchain for Trust and Transparency:** Implement immutable consultation records, advice insurance models, impact verification for donors and stakeholders, and credential verification on-chain.
- **Expansion to Livestock and Agri-Finance:** Add livestock management module, digital credit scoring, micro-insurance products, and input marketplace for supplies.
- **Offline and USSD Integration:** Develop offline mode, USSD text-based interface, SMS alerts, and voice interface for low-literacy users.

6.3 Summary

This chapter concluded the Mkulima Hub project by summarizing the achievements against each specific objective, presenting key findings from survey data and user testing, and drawing final conclusions about the platform's impact on agricultural extension services in Kenya. The survey data revealed that 58.3 percent of smallholder farmers had lost over KES 5,000 in the past year due to wrong agricultural advice, with 33.3 percent losing more than KES 20,000. These findings validated the critical need for a platform like Mkulima Hub. User testing confirmed excellent usability with a System Usability Scale score of 82.4 out of 100 and a 95 percent consultation booking success rate. Recommendations were provided for farmers, experts, administrators, and future developers.

Future work suggestions included IoT integration for soil monitoring, fine-tuning of open-source large language models for cost reduction, blockchain implementation for trust and transparency, expansion to livestock and agri-finance services, offline and USSD integration for feature phone users, and a longitudinal impact study to quantify platform benefits over multiple farming seasons. Mkulima Hub successfully demonstrated that an integrated digital ecosystem can bridge the agricultural extension service gap in Kenya, directly contributing to improved farmer decision-making, reduced post-harvest losses, and enhanced national food security.

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APPENDICES

APPENDIX A: QUESTIONNAIRE

MKULIMA HUB QUESTIONNAIRE

Collecting data for academic and development purposes. Confidentiality guaranteed.

Section 1: Respondent Profile

1. County:

☐ Mombasa	☐ Tharaka Nithi	☐ Samburu	☐ Kakamega
☐ Kwale	☐ Embu	☐ Trans Nzoia	☐ Vihiga
☐ Kilifi	☐ Kitui	☐ Uasin Gishu	☐ Bungoma
☐ Tana River	☐ Machakos	☐ Elgeyo Marakwet	☐ Busia
☐ Lamu	☐ Makueni	☐ Nandi	☐ Siaya
☐ Taita Taveta	☐ Nyandarua	☐ Baringo	☐ Kisumu
☐ Garissa	☐ Nyeri	☐ Laikipia	☐ Homa Bay
☐ Wajir	☐ Kirinyaga	☐ Nakuru	☐ Migori
☐ Mandera	☐ Murang'a	☐ Narok	☐ Kisii
☐ Marsabit	☐ Kiambu	☐ Kajiado	☐ Nyamira
☐ Isiolo	☐ Turkana	☐ Kericho	☐ Nairobi
☐ Meru	☐ West Pokot	☐ Bomet	☐ Other: ________

2. User Type: ☐ Farmer ☐ Ag-Expert ☐ Admin

3. Age Group: ☐ 18–25 ☐ 26–35 ☐ 36–45 ☐ 46–55 ☐ 56+

4. Gender: ☐ Male ☐ Female ☐ Prefer not to say

5. Device Used: ☐ Smartphone ☐ Feature Phone ☐ Borrowed ☐ None

Section 2: Problem Validation

6. How often do you experience crop losses?
☐ Never ☐ Rarely ☐ Sometimes ☐ Often ☐ Always

7. Main source of advice:
☐ Extension Officer ☐ Farmers ☐ Agrovets ☐ Media (Radio/TV) ☐ Social Media ☐ None

8. Expert interactions (last 12 months): ☐ 0 ☐ 1–2 ☐ 3–5 ☐ 6+

9. Financial loss from bad advice:
☐ > KES 20k ☐ KES 5k–20k ☐ < KES 5k ☐ No

Figure 29 Questionnaire (Page 1)

Section 3: Feature Demand – Farmers

10. Use stage-by-stage crop guides? ☐ Definitely ☐ Probably ☐ Maybe ☐ No

11. AI farming assistant usefulness (1-5): [1] [2] [3] [4] [5]

12. Willing to pay for consultation? ☐ 200 ☐ 300 ☐ 500 ☐ Maybe ☐ No

13. Top 3 booking topics:
☐ Pests ☐ Soil/Fertilizer ☐ Irrigation ☐ Seeds ☐ Prices ☐ Climate ☐ Post-Harvest

14. Join expert-moderated forum? ☐ Yes ☐ Maybe ☐ No

15. Internet reliability: ☐ Good ☐ Fair ☐ Poor ☐ None. **Offline mode?** ☐ Yes ☐ No

Section 4: Feature Demand – Agricultural Experts

16. Offer paid consultations? ☐ Yes ☐ Maybe ☐ No (Reason: ____________)

17. Format: ☐ Video ☐ Voice ☐ Text ☐ Any

18. Capacity (farmers/week): ☐ 1–5 ☐ 6–10 ☐ 11–20 ☐ 20+

19. Use impact analytics? ☐ Yes ☐ No

Section 5: Feature Demand – Administrators

20. Use monitoring dashboard? ☐ Yes ☐ Maybe ☐ No

21. Needs: ☐ User Verification ☐ Payments ☐ Impact Reports ☐ Bulk SMS ☐ Disputes

22. Incentive to switch to MkulimaHub:

- Thank You -

Figure 30 Questionnaire (Page 2)